Severity-Aware Conformal Prediction: How to Weight Probability and Loss

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Abstract

Class-conditional conformal prediction can decide which cases a human should review while bounding the share of adverse cases that pass unreviewed. The bound, however, treats all adverse cases alike, regardless of their cost. We add a constraint on the expected severity of adverse cases that pass and seek the rule that meets both constraints with the fewest reviews. When the default probability p and the expected loss given default c are known, we show that this rule is a single cut-off on the score $h_\theta = (1-\theta)\,p + \theta\,pc/B$, which blends the probability with the scaled expected loss pc. The optimal weight θ^* is unique, equals zero when severity does not bind and one when coverage does not bind, and decreases strictly in the severity level in between. We estimate the weight without labels from a plug-in dual problem and calibrate the resulting score with conformal risk control, which retains both guarantees in finite samples. The excess share of cases sent to review vanishes as the samples grow and the estimates of p and c improve. In simulations, the error of the weight estimated from the true p and c decreases like the inverse square root of the unlabeled sample size, and the excess of the full method, which also estimates p and c, falls from 11.7 to 0.5 percentage points as the samples grow. On the LendingClub loan dataset, the estimated weight follows the three regimes of the theory. At a severity level of \$60 per applicant, the method sends 74.5% of applicants to review, against 77.0% to 93.3% for four other rules that meet both levels.

1 Introduction

A lender who screens loan applications with a predictive model must decide which applications to approve automatically and which to send to a human reviewer. Conformal prediction (Vovk et al., 2005) constructs prediction sets with a distribution-free guarantee. We turn these sets into decisions. A case is approved when its set contains only the benign label, and it is sent to review when its set also contains the adverse label. Sending ambiguous sets to a human follows Singh et al. (2026). The guarantee on the sets then carries over to the decisions. Class-conditional conformal prediction (Vovk et al., 2005; Sadinle et al., 2019) bounds the share of adverse cases whose set excludes the adverse label. Such cases are *left uncovered* and pass without review.

This guarantee counts uncovered cases without regard to their severity, so a minor loss and a major one carry equal weight. Losses, however, often span several orders of magnitude, as with defaulted loans, fraudulent transfers or insurance claims. Credit risk practice accordingly measures the expected loss of a loan as the product of its probability of default and its loss given default (Basel Committee on Banking Supervision, 2006), and ranking loans by default probability alone has been found to overlook what matters to lenders (Serrano-Cinca and Gutiérrez-Nieto, 2016). On the LendingClub loan dataset, class-conditional conformal prediction at a coverage level of 10% leaves \$81.5 of default losses per applicant uncovered. This is well above our severity constraint,

which allows \$60 per applicant (Section 6).

Conformal risk control (Angelopoulos et al., 2024) can instead bound the expected severity of uncovered cases. Used with this loss alone, however, it places no bound on the number of adverse cases left uncovered, and on the same data 18% of defaults pass unreviewed. Both risks can be controlled together by calibrating one threshold per risk on a common score and adopting the most stringent (Angelopoulos et al., 2024, Proposition 6). This construction leaves open the choice of the score. Yet the score determines which cases pass, and hence how many must be reviewed to meet both levels.

Methods that control several risks take the scores as given. MultiRisk (Joshi et al., 2025) thresholds one prescribed score per risk, and CNCRC (Shen et al., 2026) treats every uncovered case as costing the largest loss. Learn-then-Test (Angelopoulos et al., 2025) controls several risks with high probability rather than in expectation. It tests a grid of thresholds and leaves part of each level unused. None of these methods asks which score should be thresholded. For a single risk, SCoRE (Bai and Jin, 2026) answers this question. It ranks cases by their estimated risk per unit reward, which selects the most cases. With two constraints, no single ratio suffices, and the best ranking depends on both levels.

The form of the optimal rule is classical. It follows from the generalized Neyman–Pearson lemma (Lehmann and Romano, 2005, Section 3.6). Charusaie and Samadi (2024) apply this lemma to learning to defer, and Tian and Feng (2025) apply the same duality to several error constraints in multi-class classification. Both estimate the multipliers from data, and their constraints hold only approximately. We use the lemma only to derive the score for coverage and severity. Our contribution lies in how the optimal weight depends on the levels, and in a calibrated rule whose guarantees hold in finite samples for any estimated score that does not depend on the calibration sample.

We address the choice of the score for coverage and severity. Our contributions are the following.

1. *The optimal weight* (Section 3). The rule with the fewest reviews thresholds a blend of the default probability and the expected loss. Its weight θ^* is unique, and it rises from zero to one as the severity level tightens (Theorem 1). The weights are monotone whenever the constraints are substitutes. This condition always holds for two constraints but can fail for more than two (Appendix A).

2. *Estimation with exact guarantees* (Section 4). The weight is estimated from unlabeled cases by a linear program, and its sampling error decreases at the rate $n_U^{-1/2}$. Calibrated by conformal risk control, the estimated score retains both guarantees in finite samples, and its excess share of cases sent to review vanishes as the samples grow and the estimates improve (Theorem 2). A weight that misses θ^* exhausts one level before the other. The share sent to review then grows in proportion to the error of the weight.

3. *Evidence* (Sections 5 and 6). Simulations with known p and c illustrate both theorems. On the LendingClub loan dataset, the estimated weight follows the three regimes of Theorem 1, and the method never sends more applicants to review than four other rules. The one exception, by 0.3 percentage points, stems from the estimated weight rather than from the form of the rule. Biased estimates of p and c place the estimated weight slightly below the best weight of one. The other rules leave part of the levels unused.

Appendix A develops the theory for any number of constraints, Appendix B contains the proofs of

the main results, and Appendix C details the simulations.

2 Problem

Setting. Each case is a triple (X, Y, C). Here $X \in \mathcal{X}$ is the information available at decision time, $Y \in \{0, 1\}$ indicates an adverse outcome, and $C \in [0, B]$ is the severity incurred if the outcome is adverse. The bound $B < \infty$ is known. A prediction set $\hat{\mathcal{C}}(x)$ is either $\{0\}$ (pass) or $\{0, 1\}$ (review). An adverse case with $1 \notin \hat{\mathcal{C}}(X)$ is left uncovered. We impose two constraints, with levels set by the operator,

$$\text{(coverage)} \quad \Pr\big(1 \notin \hat{\mathcal{C}}(X) \mid Y = 1\big) \le \alpha_{\text{cov}}, \tag{1}$$

$$\text{(severity)} \quad \mathbb{E}\big[C\,\mathbf{1}\{Y = 1,\ 1 \notin \hat{\mathcal{C}}(X)\}\big] \le \alpha_{\text{sev}}. \tag{2}$$

The first is the class-conditional guarantee for the adverse label. The second bounds the severity of uncovered cases, averaged over all cases. We write $p(x) = \Pr(Y = 1 \mid X = x)$, $c(x) = \mathbb{E}[C \mid X = x, Y = 1]$ and $\pi_1 = \Pr(Y = 1)$.

Rules. A rule $\phi : \mathcal{X} \to [0, 1]$ passes case x with probability $\phi(x)$, so $P(\phi) = \mathbb{E}[\phi(X)]$ is the share of cases passed and $1 - P(\phi)$ is the share sent to review. By the tower property (Lemma A.1 in Appendix A), the two constraints are linear in ϕ,

$$\mathbb{E}[\phi(X)\,p(X)] \le \alpha_{\text{cov}}\,\pi_1, \qquad \mathbb{E}[\phi(X)\,p(X)\,c(X)] \le \alpha_{\text{sev}}. \tag{3}$$

The first is the share of cases that are adverse and pass, and the second is the expected uncovered severity. We seek the rule that satisfies both and passes the largest share.

Conformal risk control. Conformal risk control (Angelopoulos et al., 2024) calibrates a threshold for any loss $L(t) \in [0, \bar{L}]$ that does not decrease as a cut-off t passes more cases. With losses $L_1, \ldots, L_n$ of n calibration cases,

$$\hat{t} = \sup\Big\{t \;\Big|\; \frac{1}{n+1}\Big(\textstyle\sum_{i=1}^{n} L_i(t) + \bar{L}\Big) \le \alpha\Big\} \quad \Longrightarrow \quad \mathbb{E}\big[L_{n+1}(\hat{t})\big] \le \alpha \tag{4}$$

whenever the calibration cases and a new case are exchangeable. With the indicator of a miss as the loss, it reduces to split conformal prediction. The *multi-risk CRC* applies (4) once per risk on a common score and adopts the smallest threshold. Because every loss is non-decreasing in t, each risk is then controlled at its own level.

3 The optimal weight

The rule with the fewest reviews thresholds a blend of the default probability and the expected loss, and the severity level fixes the blend. This section makes both statements precise.

The score. For prices $\lambda = (\lambda_1, \lambda_2) \ge 0$, the gain of passing x is

$$g_\lambda(x) = 1 - \lambda_1\,p(x) - \lambda_2\,p(x)\,c(x),$$

one passed case minus the use of each constraint at its price. By the generalized Neyman–Pearson lemma, a rule that passes the cases with a positive gain and uses each priced constraint in full is optimal (Proposition A.1 in Appendix A). With $S = \lambda_1 + \lambda_2 B$, the gain is positive exactly when

$$h_\theta(x) = \Big(\theta\,\frac{c(x)}{B} + 1 - \theta\Big)p(x) < \frac{1}{S}, \qquad \theta = \frac{\lambda_2 B}{S}.$$

The weight θ fixes the order of the cases, and the cut-off fixes how many pass. At $\theta = 0$ the score is the probability $p(x)$, the score of class-conditional conformal prediction. At $\theta = 1$ it is the expected loss $p(x)c(x)$ divided by B, the score of conformal risk control for severity. The division by B makes θ independent of the unit of C.

The two ends of the family. The optimal coverage-only rule is the cut-off on $p(x)$ at level α_{cov}, and S_{cc} denotes its expected uncovered severity. The optimal severity-only rule is the cut-off on $p(x)c(x)$ at level α_{sev}. Let $\underline{a}$ be the largest severity level at which the optimal severity-only rule also meets the coverage constraint. The result below requires the boundary of the passed set to be non-degenerate (Assumption A.1 in Appendix A). For coverage and severity, this requires $(p(X), p(X)c(X))$ to have a bounded density, and it fails only if $c(x)$ is constant along the boundary of the passed set.

Theorem 1 (The optimal rule and its weight)**.** *Fix $\alpha_{\text{cov}} \in (0, 1)$ and suppose Assumption A.1 holds.*

(a) An optimal rule exists. There are a weight $\theta^* \in [0, 1]$ and a cut-off $t^* > 0$ such that every optimal rule passes x if $h_{\theta^*}(x) < t^*$ and sends it to review if $h_{\theta^*}(x) > t^*$, for almost every x.

(b) The weight $\theta^*(\alpha_{\text{sev}})$ is unique, $\underline{a} \in (0, S_{\text{cc}})$, and

1. *$\theta^* = 0$ for $\alpha_{\text{sev}} \geq S_{\text{cc}}$, where the price of severity is zero,*
2. *θ^* is continuous and strictly decreasing in α_{sev} on $(\underline{a}, S_{\text{cc}})$, where both constraints bind,*
3. *$\theta^* = 1$ for $\alpha_{\text{sev}} \leq \underline{a}$, where the price of coverage is zero.*

Theorem 1 answers the question of which score to threshold. Whenever both constraints bind, the optimal ranking is given neither by the probability nor by the expected loss alone. In the two outer regimes, a single score is optimal. The two thresholds play symmetric roles. At or above S_{cc}, the optimal coverage-only rule also meets the severity constraint. At or below $\underline{a}$, the optimal severity-only rule also meets the coverage constraint. The threshold $\underline{a}$ is positive because the severity-only rule passes few defaults at a small severity level. It lies below S_{cc} because the two single-constraint rules rank applicants differently.

The proof (Appendix B.1) writes the prices as the minimizer of a convex dual function, whose Hessian is the Gram matrix G of the two losses on the boundary of the passed set. The implicit function theorem gives the derivative $-G^{-1}$ of the prices in the levels, and the sign of the off-diagonal entry of G^{-1} yields the monotonicity.

The weight has a direct reading in units of severity. Up to a positive factor, $h_\theta(x) = p(x)\,(c(x) + \kappa)$ with $\kappa = B(1 - \theta)/\theta = \lambda_1/\lambda_2$. The optimal rule therefore ranks applicants by an expected loss in which every default carries an additional fixed cost κ, the price of coverage in units of severity. This cost is infinite at $\theta = 0$, where only the number of defaults matters, and zero at $\theta = 1$, where only their losses matter. Unlike θ, it does not depend on the bound B. As the severity level tightens, κ falls, and the ranking moves from the number of defaults toward their losses. With α_{sev} fixed, loosening the coverage level also lowers κ (Proposition A.3). The two levels thus move the price of coverage in opposite directions.

Whenever both constraints bind, each single score leaves part of one level unused. By the definition

of S_{cc}, a cut-off on the probability exhausts the severity level before the coverage level. By the definition of $\underline{a}$, a cut-off on the expected loss exhausts the coverage level first. The blend uses both levels in full. Section 5 illustrates these statements on simulated data (Figure 1).

More than two constraints. The same analysis applies to any finite number of expected-loss constraints, such as coverage within groups of applicants or separate bounds on two kinds of loss (Appendix A). The optimal rule then thresholds a convex combination of the losses, and its weights are unique whenever some constraint binds. The price of each constraint never rises when its own level is loosened, a standard result of convex duality (Boyd and Vandenberghe, 2004, Section 5.6). The weights, however, are monotone only when the constraints are *substitutes*, that is, when loosening one constraint does not lower the price of another. This condition always holds for two constraints but can fail for more than two (Appendix A.3).

4 Method

4.1 Calibrated rule

By Theorem 1, the optimal rule passes a case when $h_{\theta^*}(x)$ lies below a cut-off. The calibrated rule keeps this form. It replaces p and c by estimates and chooses the cut-off on a calibration sample, so that both constraints hold in finite samples.

With estimates $\hat{p}$ and $\hat{c} \in [0, B]$ fitted on separate data, the estimated score is $\hat{h}_\theta(x) = (\theta\, \hat{c}(x)/B + 1 - \theta)\, \hat{p}(x)$. On a calibration sample of n cases, of which n_1 are adverse, the empirical risks of a cut-off t are

$$\hat{R}_{\text{cov}}(t) = \frac{1}{n_1} \sum_{i:Y_i=1} \mathbf{1}\{\hat{h}_\theta(X_i) < t\}, \qquad \hat{R}_{\text{sev}}(t) = \frac{1}{n} \sum_{i=1}^{n} C_i\, \mathbf{1}\{Y_i = 1,\ \hat{h}_\theta(X_i) < t\}.$$

The calibrated rule sets

$$\hat{t}_{\text{cov}} = \sup\Big\{ t \;\Big|\; \frac{n_1 \hat{R}_{\text{cov}}(t) + 1}{n_1 + 1} \le \alpha_{\text{cov}} \Big\}, \qquad \hat{t}_{\text{sev}} = \sup\Big\{ t \;\Big|\; \frac{n \hat{R}_{\text{sev}}(t) + B}{n + 1} \le \alpha_{\text{sev}} \Big\},$$

and passes a new case if $\hat{h}_\theta(x) < \hat{t} = \min(\hat{t}_{\text{cov}}, \hat{t}_{\text{sev}})$. The added terms 1 and B are the largest values that the two losses of the new case can take. The coverage threshold uses the adverse cases only, because coverage is conditional on $Y = 1$. The severity threshold uses all cases, because severity is defined per case.

This procedure is the multi-risk CRC applied to $\hat{h}_\theta$. Like the optimal rule, it places a single cut-off on a single score. Rules such as MultiRisk instead pass a case only when every score lies below its own cut-off, so they lack the form of the optimal rule. The shared form is what allows the calibrated rule to approach the optimal one as the estimated weight approaches θ^* (Theorem 2). Its guarantee holds in expectation, as do the constraints. We prefer it to the high-probability guarantee of Learn-then-Test, which requires a grid, a confidence level and a testing procedure to be chosen.

If the calibration cases and the new case are i.i.d., and $\hat{p}$, $\hat{c}$ and θ do not depend on the calibration sample, the calibrated rule satisfies (1) and (2) (Lemma B.1 in Appendix B). The guarantee follows from conformal risk control applied to each risk, since the smaller cut-off passes fewer cases. It holds for any estimates and any weight, which affect only the share sent to review. The remainder of this section concerns that share.

4.2 Estimating the weight without labels

The weight θ^* depends on the prices, which depend on the unknown p and c. We estimate it by solving the problem of Section 2 on an unlabeled sample $x_1, \dots, x_{n_U}$, drawn separately from the calibration sample, with $\hat{p}$ and $\hat{c}$ in place of p and c. Over $\phi = (\phi_1, \dots, \phi_{n_U}) \in [0,1]^{n_U}$, it reads

$$\max_{\phi} \sum_i \phi_i \quad \text{subject to} \quad \sum_i \phi_i \, \hat{p}(x_i) \le \alpha_{\text{cov}} \sum_i \hat{p}(x_i), \qquad \sum_i \phi_i \, \hat{p}(x_i) \, \hat{c}(x_i) \le n_U \, \alpha_{\text{sev}}.$$

This is a linear program. Its dual variables $\hat{\lambda}$ give $\hat{\theta} = \hat{\lambda}_2 B / (\hat{\lambda}_1 + \hat{\lambda}_2 B)$. No label is used, so the sample can consist of recent applicants whose outcomes are not yet known. Estimating the coefficient of a thresholded rule from unlabeled data follows Chzhen et al. (2019), who estimate in this way the threshold of a fair binary classifier. Because $\hat{\theta}$ does not depend on the calibration sample, the guarantees of the calibrated rule hold with $\hat{h}_{\hat{\theta}}$.

4.3 Accuracy of the weight and near-optimality

Let R^* be the smallest share sent to review among all rules that satisfy both constraints. Three sources can separate the calibrated rule from R^*, namely the estimates $\hat{p}$ and $\hat{c}$, the estimated weight and the finite calibration sample. Let $\tilde{\theta}$ be the weight of the population problem with $\hat{p}$ and $\hat{c}$ in place of p and c. The error of $\hat{\theta}$ then splits into a sampling error $\hat{\theta} - \tilde{\theta}$ and a bias $\tilde{\theta} - \theta^*$. We measure the accuracy of $\hat{p}$ and $\hat{c}$ by

$$e = \mathbb{E}\big[|\hat{p}(X) - p(X)| + |\hat{p}(X)\hat{c}(X) - p(X)c(X)|\big] + \alpha_{\text{cov}} \big|\mathbb{E}[\hat{p}(X) - p(X)]\big|.$$

Assumption B.1 collects three conditions for the bound on the share sent to review. Outside an event of probability ρ, $\hat{p}$ and $\hat{p}\hat{c}$ lie within η of their true values. The true score has a bounded density. Near θ^*, the share sent to review grows at most linearly as the levels tighten. Lemma B.3 shows that it also grows linearly, with slope at most H, as the weight moves away from θ^*. As in Theorem 1, all parts of the next result also require the boundary of the passed set to be non-degenerate (Assumption A.1).

Theorem 2 (Accuracy of the weight and near-optimality)**.** *Suppose Assumption A.1 holds and both constraints bind.*

(a) If Assumption A.1 also holds for $(\hat{p}(X), \hat{p}(X)\hat{c}(X))$ and both constraints bind in the problem built on $\hat{p}$ and $\hat{c}$, then $\sqrt{n_U}(\hat{\theta} - \tilde{\theta})$ is asymptotically normal, so $|\hat{\theta} - \tilde{\theta}| = O_P(n_U^{-1/2})$. With exact estimates, $|\hat{\theta} - \theta^*| = O_P(n_U^{-1/2})$.

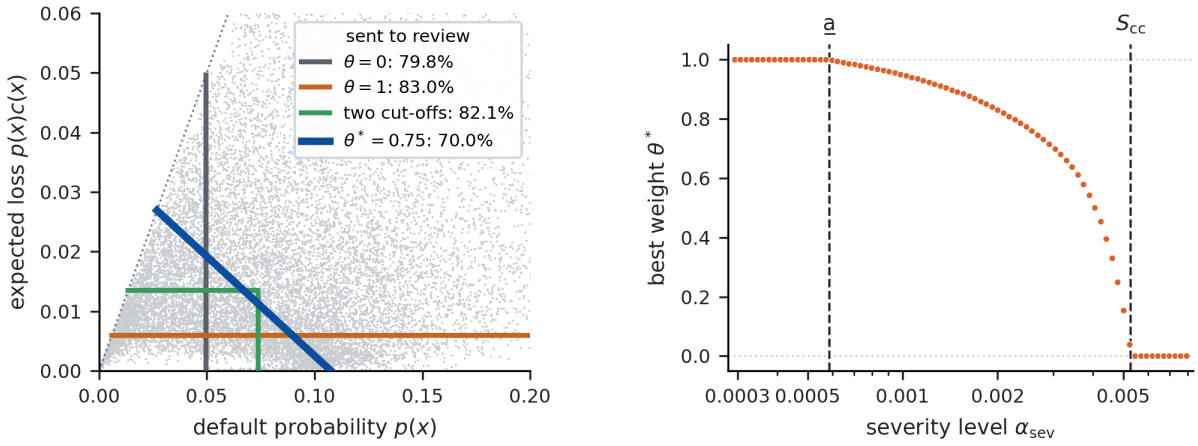
(b) If e is small enough, then $|\tilde{\theta} - \theta^*| \le C_0 \sqrt{e}$, where C_0 depends on the curvature of the dual function at its minimizer.

(c) Suppose moreover that Assumption B.1 holds and $\hat{\theta}$ lies near θ^*. For $\delta \in (0, 1/2]$, with probability at least $1 - \delta$ over the calibration sample, the calibrated rule with $\hat{h}_{\hat{\theta}}$ sends to review a share of new cases of at most

$$R^* + H|\hat{\theta} - \theta^*| + C_1\Big(\sqrt{\frac{\log(1/\delta)}{n_1}} + B\sqrt{\frac{\log(n/\delta)}{n}}\Big) + C_2\,(\eta + \rho),$$

where C_1 and C_2 depend only on the constants of Assumption B.1, on π_1 and on B.

Theorem B.1 in Appendix B.4 gives the bound with explicit constants. The rate in (a) rests on the curvature of the dual function, whose Hessian is the Gram matrix G of Section 3. The same



(a) (b)

Figure 1: Illustration of Theorem 1 on simulated data with the true p and c and $\alpha_{\text{cov}} = 0.10$. (a) Cases in the plane of $p(x)$ and $p(x)c(x)$ at $\alpha_{\text{sev}} = 0.5\, S_{\text{cc}}$, where S_{cc} is the uncovered severity of the cut-off on p at level α_{cov}. Each line bounds the cases passed by one rule, and all four rules meet both constraints. The legend gives the share each rule sends to review. No case lies above the dotted diagonal, since $c(x) \leq B$. (b) Best weight θ^* against α_{sev}, with $\underline{a}$ and S_{cc} marked by dashed lines. At each of the 80 levels, the weights within one case of the smallest share sent to review form a single interval of width at most 0.02, consistent with the uniqueness in Theorem 1.

non-degeneracy therefore yields both the monotonicity of θ^* and the rate of $\hat{\theta}$. The weight costs $H|\hat{\theta} - \theta^*|$, which is of order $n_U^{-1/2}$ with exact estimates, the same order as the calibration terms. The cost is linear rather than quadratic because the share sent to review has a kink at θ^*. A weight away from θ^* exhausts one level before the other, and the slope on each side is proportional to the price of the level left unused. The constants are not sharp, so the bound conveys rates and sources of the excess rather than its magnitude.

5 Simulation study

This section illustrates both theorems on simulated data in which, as in the LendingClub loan dataset, cases with a larger expected loss default less often (Appendix C). We set $\alpha_{\text{cov}} = 0.10$.

The optimal rule. Apart from Figure 1(b), which varies the level, we set $\alpha_{\text{sev}} = 0.5\, S_{\text{cc}}$ throughout. Since $\underline{a} = 0.11\, S_{\text{cc}}$, this level lies well inside the regime where both constraints bind. With the true p and c at this level, the blended score sends 70.0% of cases to review, against 79.8% for the probability alone, 83.0% for the expected loss alone and 82.1% for separate cut-offs on p and pc (Figure 1a). Across levels, the best weight equals one below $\underline{a}$, decreases as the level loosens, and equals zero above S_{cc}, as Theorem 1 states (Figure 1b).

Sampling error of the weight. We turn to Theorem 2 at $\alpha_{\text{sev}} = 0.5\, S_{\text{cc}}$. With the true p and c, the error of $\hat{\theta}$ falls from 0.014 at $n_U = 1{,}000$ to 0.0009 at $n_U = 256{,}000$ (Figure 2a). Its logarithm decreases with slope -0.51 in $\log n_U$, close to the rate $n_U^{-1/2}$ of Theorem 2(a).

Bias from the estimates. To isolate the bias of Theorem 2(b), we distort the true p or c and compute the weight $\tilde{\theta}$ of the population problem built on the distorted values. Compared at the

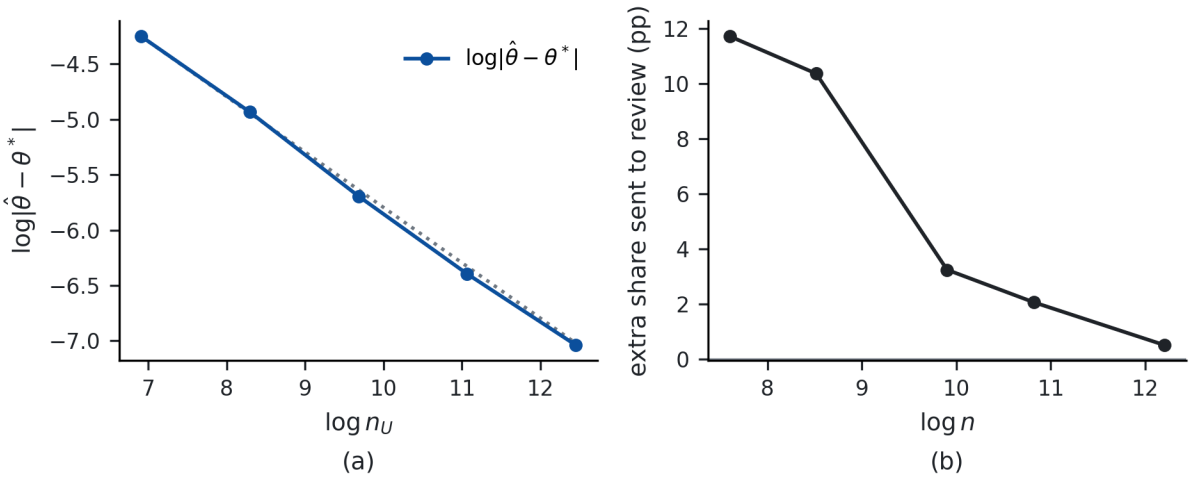


Figure 2: Illustration of Theorem 2 on the simulated data of Figure 1, with $\alpha_{\text{cov}} = 0.10$ and $\alpha_{\text{sev}} = 0.5\, S_{\text{cc}}$. Logarithms are natural. (a) Mean error of the weight estimated from the true p and c on n_U unlabeled cases, over 100 repetitions. An error proportional to $n_U^{-1/2}$ appears as a line of slope $-1/2$, shown dotted. (b) Extra share sent to review by the full method over the optimal rule, against the size n of each of the training, unlabeled and calibration samples. Means over 20 repetitions. The standard error is at most 1.6 percentage points at $n = 2{,}000$ and at most 0.52 for $n \geq 20{,}000$.

same error e, errors in the expected loss given default move the weight about 1.6 times as much as errors in the probability. The direction of an error also matters. Pushing the probabilities toward zero and one shifts the weight more than pulling them toward one half, even at a smaller error e. The gap grows with the distortion and reaches a factor of about three when the logit is scaled by 1.2 rather than 0.8. Across all twenty distortions, ten scalings of c and ten scalings of the logit of p by factors from 0.8 to 1.2, the bias shrinks roughly like $e^{0.83}$ as e falls. It therefore vanishes faster than the bound $C_0\sqrt{e}$ of Theorem 2(b). The bound holds for every error of size e, and the worst such errors move the weight by order $\sqrt{e}$. Smooth distortions such as these move it only in proportion to e.

The full method. The full method estimates p and c on a training sample, estimates the weight and calibrates the cut-off (Appendix C). Its share sent to review exceeds the optimal share of 70.1% by 11.7 percentage points at $n = 2{,}000$, by 3.2 at $n = 20{,}000$ and by 0.5 at $n = 200{,}000$, as Theorem 2(c) predicts (Figure 2b). Both guarantees hold at every n. The share of defaults left uncovered lies between 0.067 and 0.100, and the uncovered severity stays below α_{sev}.

6 Experiments on the LendingClub loan dataset

The simulations illustrate the theorems with known p and c. On real data these are unknown, so the experiments address four questions. Are the probability and the loss misaligned, so that the problem arises in practice? Do the guarantees hold? Does the weight estimated without labels follow the three regimes of Theorem 1? How many reviews do the other rules spend on the part of the levels they leave unused?

Data and models. We use the terminated 36-month loans issued from 2012 to 2015 in the LendingClub loan dataset (589,488 loans, 29 origination features). An adverse case is a default, and its severity is the dollar loss net of recoveries, bounded by the maximum loan amount $B = \$35{,}000$.

On loans issued before 2015, we fit a LightGBM classifier for $\hat{p}$ and a LightGBM regressor for the loss rate of defaults, holding out a random 10% for validation. The expected loss given default $\hat{c}$ is the predicted loss rate times the loan amount. Appendix D gives the details.

Levels and weight. We set $\alpha_{\text{cov}} = 0.10$ and vary α_{sev} over 20 levels spaced evenly on a log scale from \$5 to \$100 per applicant, together with \$60, the level of Table 1. This range covers the three regimes of Theorem 1, since class-conditional conformal prediction leaves \$64.5 per applicant uncovered on the validation loans. At each level, $\hat{\theta}$ is computed on the 30,646 validation loans without their outcomes. Table 1 uses $\alpha_{\text{sev}} = \$60$, which is 10% of the default loss per applicant on the validation loans and thus matches the coverage level of 10%.

Calibration and evaluation. The 283,026 loans issued in 2015 are split at random into calibration and test halves 100 times. We report means over the splits of the share of defaults left uncovered, the uncovered loss per applicant, the share of high-loss defaults left uncovered and the share of applicants sent to review. A default is high-loss if its loss is at least \$10,607, the 90th percentile of validation default losses.

Compared rules. All compared rules satisfy both constraints. The proposed method is the multi-risk CRC on $\hat{h}_{\hat{\theta}}$. The same calibration on $\hat{p}$ alone and on $\hat{p}\hat{c}$ alone gives the two ends of the score family, $\theta = 0$ and $\theta = 1$. The separate rule calibrates a coverage threshold on $\hat{p}$ and a severity threshold on $\hat{p}\hat{c}$ and passes a case only below both. It has the form of MultiRisk with one score per risk, although MultiRisk sets its thresholds in sequence rather than separately. Following CNCRC (Shen et al., 2026), the worst-case rule treats every uncovered default as costing B and calibrates $\Pr(Y = 1, 1 \notin \hat{\mathcal{C}}(X)) \leq \alpha_{\text{sev}}/B$ together with coverage on $\hat{p}$.

Misaligned probability and loss. The largest fifth of loans by amount defaults slightly less often than the average loan, 13.3% against 14.9% among 2015 loans. Its loss per default, however, exceeds that of the smallest fifth by a factor of more than six. A ranking by probability alone therefore leaves the largest losses uncovered first. Class-conditional conformal prediction at $\alpha_{\text{cov}} = 0.10$ leaves \$81.5 per applicant uncovered, above the severity level of \$60 used in Table 1. The severity-only rule at $\alpha_{\text{sev}} = \$60$ meets that level but leaves 18.3% of defaults uncovered, nearly twice the coverage level.

Guarantees. Across the 21 levels and the five rules, the share of defaults left uncovered ranges from 0.001 to 0.101, and the uncovered loss reaches at most 100.7% of its level. The largest excess over a level is 2.0 standard errors over the 100 splits for coverage and 1.3 for severity, which is consistent with sampling noise over 105 comparisons. Both guarantees hold in expectation, so a mean at the level within such noise is expected. They also hold whatever weight is used, so an error in $\hat{\theta}$ costs reviews but never coverage or severity.

Why calibrate. A plug-in rule could skip calibration. It uses the same weight $\hat{\theta}$ but sets the cut-off from $\hat{p}$ and $\hat{c}$ on the calibration loans, without their outcomes. At \$60 it sends 72.6% of applicants to review, 1.9 percentage points fewer than the proposed rule, and its uncovered loss reaches 114% of the severity level. Over the 21 levels, it exceeds the severity level at 19, by up to 20%, because the estimates understate losses in 2015. Calibration with outcomes gives up these reviews to keep both guarantees.

The estimated weight. The estimated weight follows the three regimes of Theorem 1 (Figure 3a). It equals one up to \$21, where only severity binds. It decreases from 0.996 at \$24 to 0.489 at \$62, where both constraints bind. It equals zero from \$73, above the uncovered loss of \$64.5 left

	Sent to review (%)	Coverage level used (%)	Severity level used (%)	Uncovered defaults (share)	Loss from uncovered ($ per app.)	High-loss uncovered (%)
Proposed, $\hat{\theta}$	74.5	85	100	0.085	60.2	7.8
Probability only, $\theta = 0$	77.0	73	100	0.073	60.3	9.3
Expected loss only, $\theta = 1$	79.9	101	46	0.101	27.9	0.5
Separate cut-offs	81.9	59	51	0.059	30.5	1.8
Worst-case cost (CNCRC)	93.3	12	17	0.012	10.3	1.7

Table 1: Rules at the same levels, $\alpha_{\text{cov}} = 0.10$ and $\alpha_{\text{sev}} = \$60$, on the 2015 loans of the LendingClub loan dataset. Means over 100 random splits. The standard error is at most 0.05 percentage points for the share sent to review and at most $0.32 for the uncovered loss. The second and third columns give the share of each level that the rule uses.

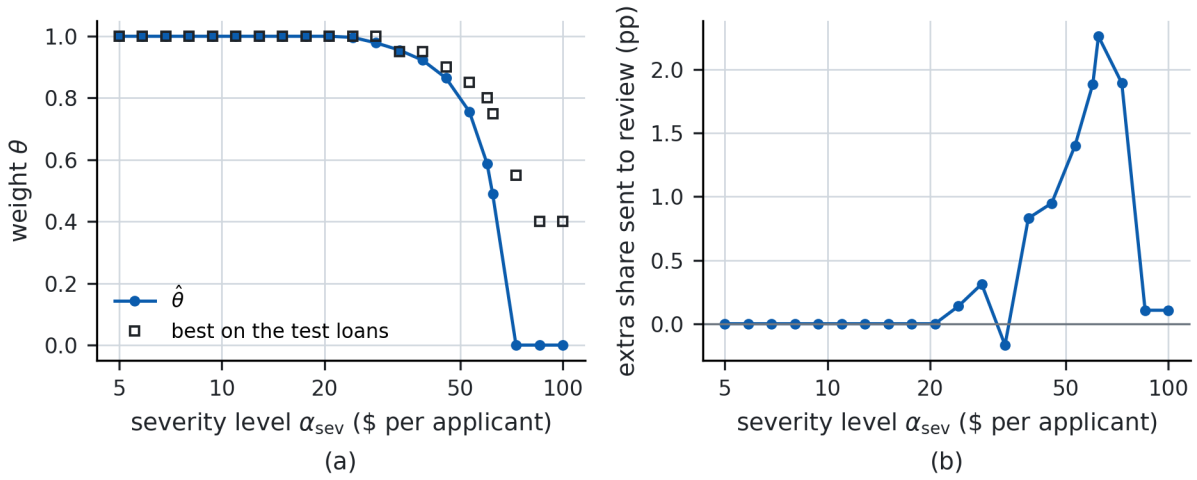


Figure 3: The weight on the 2015 loans of the LendingClub loan dataset, with $\alpha_{\text{cov}} = 0.10$. (a) The weight $\hat{\theta}$ estimated on the validation loans without their outcomes, and the grid weight with the smallest share sent to review on the test halves among the weights $0, 0.05, \ldots, 1$, which uses the test outcomes. The axis of α_{sev} is on a log scale. (b) Share sent to review with $\hat{\theta}$ minus the share with the best weight on the test loans. Means over 100 random splits.

by class-conditional conformal prediction on the validation loans. The price of coverage κ rises over the middle regime from $123 to $36,557 per default. At $60 it equals $24,542, so the score ranks applicants as if every default cost this amount on top of its loss. Below $21 the estimated weight coincides with the best weight on the test loans. This is the grid weight that sends the fewest applicants to review on the test halves. It uses the test outcomes and is therefore not available in practice.

The gap to the best weight. The proposed rule does not use both levels in full. At $60 it uses all of the severity level but 85% of the coverage level (Table 1). In the middle regime its weight lies below the best weight on the test loans, by 0.21 at $60 and 0.55 at $73, and it sends up to 2.1 percentage points more applicants to review (Figure 3b). The extra share sent to review grows with the level for two reasons. First, the optimal weight changes slowly near one and steeply near the upper end of the middle regime, so the same bias in $\hat{p}$ and $\hat{c}$ moves the weight more at looser levels. Second, by Lemma B.3, a weight below the optimum exhausts severity first and leaves coverage unused, so each unit of error costs reviews in proportion to the price of coverage. This price rises

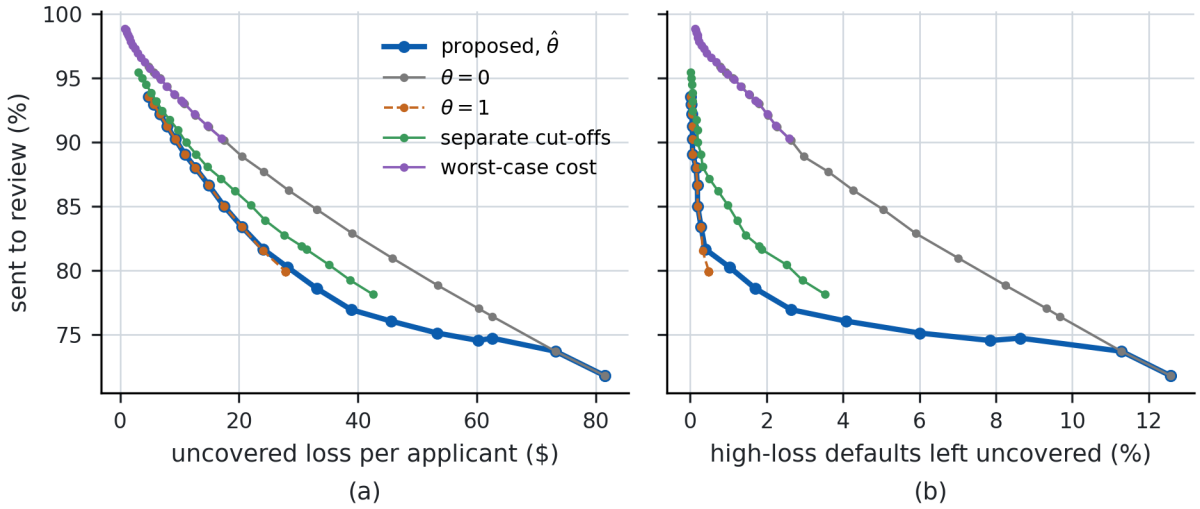


Figure 4: Share of applicants sent to review against (a) the uncovered loss per applicant and (b) the share of high-loss defaults left uncovered, on the 2015 loans of the LendingClub loan dataset. Each curve traces one rule as α_{sev} varies from \$5 to \$100 with $\alpha_{\text{cov}} = 0.10$. Means over 100 random splits into calibration and test halves.

with the level. Above \$73 the extra share nearly vanishes, since severity barely binds on the 2015 loans and the share sent to review hardly depends on the weight.

The bias of $\hat{p}$ and $\hat{c}$ appears only on the 2015 loans. On the validation loans, the mean of $\hat{p}$ is 13.3% against a default rate of 13.1%, and the mean of $\hat{p}\hat{c}$ is \$592 against a loss of \$600 per applicant. On the 2015 loans, these pairs are 12.8% against 14.9% and \$577 against \$730. The model thus underestimates both defaults and losses in 2015, and $\hat{\theta}$ targets the weight that is optimal under $\hat{p}$ and $\hat{c}$ rather than under the 2015 outcomes, as in Theorem 2(b). Since actual losses are larger, a given severity level is tighter on the 2015 loans than the estimates suggest. By Theorem 1, a tighter level calls for a larger weight, so the estimated weight falls below the best one. Since the weight requires no outcomes, one might instead estimate it on recent applicants. This does not help, because the shift between periods lies in the outcomes rather than in the features. Computed from $\hat{p}$ and $\hat{c}$ alone, the uncovered loss of class-conditional conformal prediction is \$69.0 on the validation loans and \$66.4 on the 2015 loans, whereas the observed value rises from \$64.5 to \$81.5. Estimated on a separate third of the 2015 loans without their outcomes, the weight leaves a slightly larger gap, 2.5 instead of 2.0 percentage points at \$60. Re-estimating $\hat{p}$ and $\hat{c}$ on recent applicants could reduce the bias, but it requires their outcomes, which for 36-month loans arrive only years later.

The cost of fewer reviews. Table 1 compares the rules at $\alpha_{\text{sev}} = \$60$. The proposed rule sends 74.5% of applicants to review, against 77.0% to 93.3% for the other rules. The second and third columns show where the other reviews go. The rule with $\theta = 1$ uses 46% of the severity level, the separate cut-offs use about half of each level, and the worst-case rule uses less than a fifth of each. Each of these rules thus spends reviews on protection beyond what the levels require, and the proposed rule converts this unused part into fewer reviews. The price is the protection given up. The proposed rule leaves \$60.2 per applicant uncovered against \$27.9 for $\theta = 1$, and 7.8% of high-loss defaults against 0.5%. A lender who sets the level at \$60 accepts this loss, and one who does not should set a lower level.

Reviews at equal protection. Figure 4 compares the rules at equal protection over all levels. At levels up to \$21 the estimated weight equals one, so the proposed rule and the rule with $\theta = 1$ coincide. At higher levels the rule with $\theta = 1$ stops at 79.9% of applicants sent to review. Its ranking by expected loss exhausts the coverage level, so a looser severity level cannot pass more applicants. The proposed rule continues down to 71.8%. The separate cut-offs send more applicants to review at every level of protection, and the worst-case rule sends far more. The rule with $\theta = 0$ sends more until \$73, from where $\hat{\theta} = 0$ and the two rules coincide. Over all 21 levels, the proposed rule sends on average 3.5, 2.2, 5.4 and 13.4 percentage points fewer applicants to review than the rules with $\theta = 0$ and $\theta = 1$, the separate cut-offs and the worst-case rule. It sends more than one of them at a single level, by 0.3 points against $\theta = 1$ at \$28. There the best weight on the test loans is one, while the biased estimates place $\hat{\theta}$ at 0.979. For high-loss defaults, however, the rule with $\theta = 1$ leaves slightly fewer uncovered than the proposed rule at a similar share sent to review. The difference appears only at the levels \$24 and \$28, where about 80% of applicants are sent to review (Figure 4b). The rule with $\theta = 1$ does better there because it passes high-loss defaults last, a criterion that neither constraint imposes.

7 Discussion

We asked which score a review rule should threshold to control coverage and severity with the fewest reviews. The answer is a blend of the default probability and the expected loss, whose weight rises as the severity level tightens. The weight can be estimated without labels, the calibrated rule retains both guarantees in finite samples, and its excess share sent to review vanishes as the samples grow and the estimates of p and c improve. On the LendingClub loan dataset, the estimated weight follows the three regimes of the theory. The method sends fewer applicants to review than rules that fix the score or calibrate one score per risk, because those rules leave part of the levels unused. Its weight falls below the best weight on the test loans, at a cost of up to 2.1 percentage points, because the model underestimates defaults and losses in the later period.

Limitations of the theory. The bound of Theorem 2(c) rests on high-level conditions, and conditions on the distribution of $(p(X), p(X)c(X))$ that imply them are not derived. The bias bound $C_0\sqrt{e}$ is pessimistic. In the simulation of Section 5 the bias shrinks roughly like $e^{0.83}$. Where severity just stops binding, $\theta^* = 0$ lies on the edge of its domain and the limit law of $\hat{\theta}$ is not normal (Self and Liang, 1987; Andrews, 1999). The results for more than two constraints are supported by proofs only, and the possible failure of weight monotonicity for more than two constraints is shown by an example with three. Section 6 suggests a third constraint on high-loss defaults, which neither level controls. With three constraints, however, the weights need not be monotone in the levels.

Limitations of the evidence. Outcomes exist only for approved loans, which form a selected and safer part of all applicants. Approval usually follows a credit-score cut-off, so approved and rejected applicants likely overlap little, and reweighting such as weighted conformal prediction (Tibshirani et al., 2019) cannot extend the guarantees to all applicants. Both guarantees hold in expectation over the calibration draw, and Learn-then-Test gives a high-probability version at a higher share sent to review. They also require the calibration loans and the new loans to be exchangeable. In deployment, calibration must use loans whose outcomes are already known, and these loans are older than the new applicants. Our random split of the 2015 loans ignores this delay. Calibrated instead on half of the validation loans, issued before 2015, the proposed rule exceeds the severity level at 20 of the 21 levels and the coverage level at 12, by up to 44% and 15%. At \$60 it uses 128% of the severity level and 110% of the coverage level. The rules with $\theta = 0$ and $\theta = 1$ exceed the levels as well, so the failure stems from the shift between periods rather than from the weight. The weight inherits any bias of $\hat{p}$ and $\hat{c}$, and re-estimating them requires recent outcomes

(Section 6). Finally, the real-data results come from one dataset with losses capped at the loan amount. Heavier tails, as in fraud or insurance, remain untested.

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A Theory for any number of constraints

This appendix states the problem for K expected-loss constraints, characterizes its optimal rule and studies how the prices and the weights depend on the levels. Theorem 1 is the case $K = 2$ with coverage and severity, and its proof in Appendix B.1 draws on the results below.

A.1 Setting and the optimal rule

Constraints. For $k = 1, \dots, K$, let $L_k \geq 0$ be a loss incurred only when a case passes, with conditional mean $\ell_k(x) = \mathbb{E}[L_k \mid X = x]$ and level $a_k > 0$. A rule ϕ satisfies constraint k if $\mathbb{E}[\phi(X)\,\ell_k(X)] \leq a_k$. Coverage and severity are the case $K = 2$ with $\ell_1(x) = p(x)$, $a_1 = \alpha_{\text{cov}}\pi_1$, $\ell_2(x) = p(x)c(x)$ and $a_2 = \alpha_{\text{sev}}$, by the following lemma. Larger K arises when coverage is required separately within groups of applicants, with $\ell_k(x) = p(x)\mathbf{1}\{x \in \text{group } k\}$. It also arises when two kinds of loss, such as the unrecovered principal and the cost of recovery, each have a severity bound.

Lemma A.1. *For any $\phi : \mathcal{X} \to [0, 1]$,*

$$\mathbb{E}\big[C\,\mathbf{1}\{Y = 1\}\,\phi(X)\big] = \mathbb{E}\big[\phi(X)\,p(X)\,c(X)\big], \qquad \mathbb{E}\big[\mathbf{1}\{Y = 1\}\,\phi(X)\big] = \mathbb{E}\big[\phi(X)\,p(X)\big].$$

If a rule passes x with probability $\phi(x)$, that is, $\mathbf{1}\{1 \notin \hat{\mathcal{C}}(X)\} = \mathbf{1}\{U \leq \phi(X)\}$ with $U \sim \text{Unif}(0, 1)$ independent of (X, Y, C), then the constraints (1) and (2) are equivalent to (3).

Proof. By the tower property,

$$\begin{aligned}\mathbb{E}\big[C\,\mathbf{1}\{Y = 1\}\,\phi(X) \mid X\big] &= \phi(X)\,\mathbb{E}\big[\mathbf{1}\{Y = 1\}\,\mathbb{E}[C \mid X, Y] \mid X\big] \\ &= \phi(X)\,\Pr(Y = 1 \mid X)\,\mathbb{E}[C \mid X, Y = 1] = \phi(X)\,p(X)\,c(X).\end{aligned}$$

Taking expectations gives the first identity, and $C \equiv 1$ gives the second. Since U is independent of (X, Y, C), $\Pr(U \leq \phi(X) \mid X, Y, C) = \phi(X)$, so the uncovered severity equals $\mathbb{E}[C\,\mathbf{1}\{Y = 1\}\,\phi(X)]$ and $\Pr(Y = 1,\ 1 \notin \hat{\mathcal{C}}(X)) = \mathbb{E}[\phi(X)\,p(X)]$. Dividing the latter by π_1 gives the coverage constraint. □

Rules and the feasible set. Rules are elements of

$$\Phi = \big\{\phi \in L_\infty(P_X) \;\big|\; 0 \leq \phi(x) \leq 1 \text{ for almost every } x\big\},$$

where $L_\infty(P_X)$ is the space of functions on $\mathcal{X}$ that are bounded with probability one under the distribution P_X of X, and two functions that differ only on a set of probability zero are identified. For levels $a = (a_1, \dots, a_K)$, the feasible set and the largest share passed are

$$\Phi_a = \big\{\phi \in \Phi \;\big|\; \mathbb{E}[\phi(X)\,\ell_k(X)] \leq a_k \text{ for } k = 1, \dots, K\big\}, \qquad P^* = \sup_{\phi \in \Phi_a} P(\phi).$$

For $\lambda \in [0, \infty)^K$, the gain of passing x and the value of a rule against the gains are

$$g_\lambda(x) = 1 - \textstyle\sum_k \lambda_k \ell_k(x), \qquad N_\lambda(\phi) = \mathbb{E}[\phi(X)\,g_\lambda(X)].$$

For $\lambda \in [0, \infty)^K$, a rule $\phi^* \in \Phi$ has the *form of the optimal rule* if

$$\phi^*(x) = 1 \;\text{ if } g_\lambda(x) > 0, \qquad \phi^*(x) = 0 \;\text{ if } g_\lambda(x) < 0. \tag{5}$$

Proposition A.1 (Optimal rule)**.** *Suppose that every $\ell_k(X)$ is integrable and every $a_k > 0$.*

(a) Let $\lambda \in [0,\infty)^K$, and let ϕ^* pass x if $g_\lambda(x) > 0$ and review it if $g_\lambda(x) < 0$. If $\phi^* \in \Phi_a$ and $\mathbb{E}[\phi^*(X)\,\ell_k(X)] = a_k$ for every k with $\lambda_k > 0$, then $P(\phi^*) = P^*$.

(b) An optimal rule exists. Moreover, there is $\lambda \in [0,\infty)^K$ such that every optimal rule has the form in (a) wherever $g_\lambda(x) \neq 0$ and uses every constraint with $\lambda_k > 0$ in full.

The score. The prices carry both the order of the cases and the position of the cut-off, and the two can be separated. Fix scales $s \in (0,\infty)^K$ that put the losses on a common unit, for example upper bounds on ℓ_k. For $\lambda \neq 0$, let $S_\lambda = \sum_j s_j\lambda_j$ and define the weights and the score

$$w_k(\lambda) = \frac{s_k\lambda_k}{S_\lambda}, \qquad h_w(x) = \sum_{k=1}^{K} w_k\,\frac{\ell_k(x)}{s_k}, \qquad w \in \Delta^{K-1},$$

where Δ^{K-1} is the probability simplex. Then $\sum_k \lambda_k\ell_k(x) = S_\lambda\, h_{w(\lambda)}(x)$, so the optimal rule passes the cases with $h_w(x) < 1/S_\lambda$. For coverage and severity with $s = (1, B)$, the weights are $w = (1-\theta, \theta)$ and the score is h_θ of Section 3.

The proof decomposes the share passed into the expected gain and the priced use of the constraints. The rule in (a) maximizes the expected gain pointwise, and complementary slackness brings the priced use to its largest feasible value. The prices in (b) are obtained from a separating hyperplane.

Two facts are used in both parts.

Lemma A.2 (Decomposition and pointwise bound)**.** *Let $\lambda \in [0,\infty)^K$ and let $\phi^* \in \Phi$ satisfy (5). For every $\phi \in \Phi$,*

(i) $P(\phi) = N_\lambda(\phi) + \sum_k \lambda_k\mathbb{E}[\phi(X)\,\ell_k(X)]$,

(ii) $N_\lambda(\phi) \le N_\lambda(\phi^*)$, with equality if and only if $\phi(x) = \phi^*(x)$ for almost every x with $g_\lambda(x) \neq 0$.

Proof. (i) At every x,

$$\phi(x) = \phi(x)\Big(g_\lambda(x) + \textstyle\sum_k \lambda_k\ell_k(x)\Big) = \phi(x)\,g_\lambda(x) + \sum_k \lambda_k\,\phi(x)\,\ell_k(x).$$

Each term on the right is integrable, since $|\phi(x)\,\ell_k(x)| \le |\ell_k(x)|$ and $|g_\lambda(x)| \le 1 + \sum_k \lambda_k|\ell_k(x)|$. Taking expectations gives (i).

(ii) Let $\Delta(x) = \big(\phi^*(x) - \phi(x)\big)g_\lambda(x)$. If $g_\lambda(x) > 0$, then $\phi^*(x) = 1 \ge \phi(x)$. If $g_\lambda(x) < 0$, then $\phi^*(x) = 0 \le \phi(x)$. If $g_\lambda(x) = 0$, then $\Delta(x) = 0$. Hence $\Delta(x) \ge 0$ at every x, and

$$N_\lambda(\phi^*) - N_\lambda(\phi) = \mathbb{E}[\Delta(X)] \ge 0.$$

A non-negative random variable has zero mean if and only if it is zero with probability one. So equality holds if and only if $\Delta(x) = 0$ for almost every x, which for $g_\lambda(x) \neq 0$ means $\phi(x) = \phi^*(x)$. □

Proof of Proposition A.1(a). Let $\phi \in \Phi_a$. For every k, $\lambda_k \ge 0$ and $\mathbb{E}[\phi(X)\,\ell_k(X)] \le a_k$, so

$$\lambda_k\,\mathbb{E}[\phi(X)\,\ell_k(X)] \le \lambda_k a_k. \tag{6}$$

For ϕ^* the same terms are equalities,

$$\lambda_k\big(\mathbb{E}[\phi^*(X)\,\ell_k(X)] - a_k\big) = 0 \quad \text{for every } k, \tag{7}$$

because either $\lambda_k = 0$, or $\lambda_k > 0$ and the bracket is zero by hypothesis. Condition (7) is called complementary slackness. Then

$$\begin{aligned}
P(\phi) &= N_\lambda(\phi) + \textstyle\sum_k \lambda_k \mathbb{E}[\phi(X)\,\ell_k(X)] && \text{by Lemma A.2(i)}\\
&\le N_\lambda(\phi^*) + \textstyle\sum_k \lambda_k \mathbb{E}[\phi(X)\,\ell_k(X)] && \text{by Lemma A.2(ii)}\\
&\le N_\lambda(\phi^*) + \textstyle\sum_k \lambda_k a_k && \text{by (6)}\\
&= N_\lambda(\phi^*) + \textstyle\sum_k \lambda_k \mathbb{E}[\phi^*(X)\,\ell_k(X)] && \text{by (7)}\\
&= P(\phi^*) && \text{by Lemma A.2(i).}
\end{aligned}$$

Since $\phi^* \in \Phi_a$, this gives $P(\phi^*) = \sup_{\phi\in\Phi_a} P(\phi) = P^*$. □

Proof of Proposition A.1(b). (b1) An optimal rule exists. The space $L_\infty(P_X)$ is the dual of $L_1(P_X)$, and its weak-* topology is the weakest topology in which $\phi \mapsto \mathbb{E}[\phi(X)\,h(X)]$ is continuous for every integrable h. By the Banach–Alaoglu theorem, the unit ball $\{\phi \mid \|\phi\|_\infty \le 1\}$ is weak-* compact. The set Φ_a is the intersection of the unit ball with the sets

$$\big\{\phi \,\big|\, \mathbb{E}[\phi(X)\,h(X)] \ge 0\big\}, \quad \big\{\phi \,\big|\, \mathbb{E}[(1-\phi(X))\,h(X)] \ge 0\big\} \quad \text{for every integrable } h \ge 0,$$

$$\big\{\phi \,\big|\, \mathbb{E}[\phi(X)\,\ell_k(X)] \le a_k\big\} \quad \text{for } k = 1, \dots, K,$$

which express $\phi \ge 0$, $\phi \le 1$ and the constraints. Each is the preimage of a closed half-line under a weak-* continuous map, because $h(X)$, $\ell_k(X)$ and 1 are integrable. Hence Φ_a is weak-* closed, and as a closed subset of a compact set it is compact. It is non-empty, since $\phi \equiv 0$ belongs to it. The map $\phi \mapsto P(\phi) = \mathbb{E}[\phi(X)\cdot 1]$ is weak-* continuous, and a continuous function on a non-empty compact set attains its maximum. So $P(\phi^\circ) = P^*$ for some $\phi^\circ \in \Phi_a$.

(b2) The multipliers exist. Define the sets of achievable and of better-than-optimal pairs in $\mathbb{R}^K \times \mathbb{R}$,

$$\begin{aligned}
\mathsf{A} &= \big\{(u,r) \,\big|\, \text{there is } \phi \in \Phi \text{ with } \mathbb{E}[\phi(X)\,\ell_k(X)] \le u_k \text{ for every } k \text{ and } P(\phi) \ge r\big\},\\
\mathsf{B} &= \big\{(u,r) \,\big|\, u_k < a_k \text{ for every } k \text{ and } r > P^*\big\}.
\end{aligned}$$

If ϕ and ϕ' witness (u,r) and (u',r') and $t \in [0,1]$, then $t\phi + (1-t)\phi' \in \Phi$ and, by linearity of the expectation, it witnesses $t(u,r) + (1-t)(u',r')$. So A is convex. The set B is convex, open and non-empty. A pair in both would be witnessed by some ϕ with $\mathbb{E}[\phi(X)\,\ell_k(X)] < a_k$ for every k and $P(\phi) > P^*$, which contradicts the definition of P^*. So the two sets are disjoint. By the separating hyperplane theorem in $\mathbb{R}^{K+1}$, there are $(\mu,\nu) \in \mathbb{R}^K \times \mathbb{R}$, not both zero, and $b \in \mathbb{R}$ with

$$\nu r - \mu\cdot u \le b \;\text{ for } (u,r) \in \mathsf{A}, \qquad \nu r - \mu\cdot u \ge b \;\text{ for } (u,r) \in \mathsf{B}. \tag{8}$$

The signs follow from the shape of A. If $(u,r) \in \mathsf{A}$, then $(u + se_k, r) \in \mathsf{A}$ and $(u, r-s) \in \mathsf{A}$ for every $s \ge 0$, where e_k is the k-th unit vector. By (8),

$$\nu r - \mu\cdot u - s\mu_k \le b \quad \text{and} \quad \nu r - \mu\cdot u - s\nu \le b \qquad \text{for every } s \ge 0,$$

which is possible only if $\mu_k \ge 0$ and $\nu \ge 0$.

Next, $\nu > 0$. Suppose $\nu = 0$. The rule $\phi \equiv 0$ gives $(0,0) \in \mathsf{A}$, so $b \ge 0$. The pair $(a/2, P^*+1)$ lies in B, so $-\mu\cdot a/2 \ge b \ge 0$. Since $\mu_k \ge 0$ and $a_k > 0$ for every k, this forces $\mu = 0$, and then $(\mu,\nu) = 0$, a contradiction.

Let $\lambda = \mu/\nu \in [0,\infty)^K$ and $b' = b/\nu$. Dividing (8) by ν,

$$r - \lambda \cdot u \le b' \ \text{ on } \mathsf{A}, \qquad r - \lambda \cdot u \ge b' \ \text{ on } \mathsf{B}.$$

The pairs $(a - \varepsilon\mathbf{1}, P^* + \varepsilon)$ lie in B for $\varepsilon > 0$, and letting $\varepsilon \downarrow 0$ gives $P^* - \lambda \cdot a \ge b'$. Every $\phi \in \Phi$ gives the pair $\big((\mathbb{E}[\phi(X)\,\ell_k(X)])_k,\ P(\phi)\big) \in \mathsf{A}$, so by Lemma A.2(i)

$$N_\lambda(\phi) + \textstyle\sum_k \lambda_k a_k = P(\phi) - \sum_k \lambda_k \mathbb{E}[\phi(X)\,\ell_k(X)] + \sum_k \lambda_k a_k \le b' + \lambda \cdot a \le P^* \qquad \text{for every } \phi \in \Phi. \tag{9}$$

The multipliers λ are constructed without reference to any particular optimal rule.

(b3) Form and complementary slackness. Let $\phi^\circ \in \Phi_a$ with $P(\phi^\circ) = P^*$, and let $\phi^* \in \Phi$ satisfy (5) with the λ of (b2), for example $\phi^*(x) = \mathbf{1}\{g_\lambda(x) > 0\}$. Then

$$\begin{aligned}
P^* = P(\phi^\circ) &= N_\lambda(\phi^\circ) + \textstyle\sum_k \lambda_k \mathbb{E}[\phi^\circ(X)\,\ell_k(X)] && \text{by Lemma A.2(i)} \\
&\le N_\lambda(\phi^\circ) + \textstyle\sum_k \lambda_k a_k && \text{by (6), since } \phi^\circ \in \Phi_a \\
&\le N_\lambda(\phi^*) + \textstyle\sum_k \lambda_k a_k && \text{by Lemma A.2(ii)} \\
&\le P^* && \text{by (9) with } \phi = \phi^*.
\end{aligned}$$

The first and last terms are equal, so every inequality is an equality. Equality in the third line gives $N_\lambda(\phi^\circ) = N_\lambda(\phi^*)$, and Lemma A.2(ii) shows that ϕ° satisfies (5) for almost every x with $g_\lambda(x) \ne 0$. Equality in the second line gives

$$\textstyle\sum_k \lambda_k \big(\mathbb{E}[\phi^\circ(X)\,\ell_k(X)] - a_k\big) = 0.$$

Every term of this sum is at most zero by (6), so every term is zero. Hence $\lambda_k > 0$ implies $\mathbb{E}[\phi^\circ(X)\,\ell_k(X)] = a_k$. □

A.2 The dual function and the prices

Write $Z = (\ell_1(X), \dots, \ell_K(X))$, so that $g_\lambda(X) = 1 - \lambda \cdot Z$, and define the dual function

$$D(\lambda; a) = \mathbb{E}\big[\max(0,\ 1 - \lambda \cdot Z)\big] + \lambda \cdot a.$$

Write $P^*(a)$ for the largest share passed at levels a, and let $\Lambda^*(a)$ be the set of minimizers of $D(\cdot\,; a)$ over $[0,\infty)^K$.

Lemma A.3 (The dual function)**.** *Under the conditions of Proposition A.1,*

(i) $P(\phi) \le D(\lambda; a)$ *for every* $\phi \in \Phi_a$ *and every* $\lambda \in [0,\infty)^K$,

(ii) $P^*(a) = \min_{\lambda \ge 0} D(\lambda; a)$, *and* $\Lambda^*(a)$ *contains the multiplier of the proof of Proposition A.1(b),*

(iii) every $\lambda \in \Lambda^*(a)$ *has the properties stated in Proposition A.1(b).*

Proof. (i) Let $\phi^*(x) = \mathbf{1}\{g_\lambda(x) > 0\}$. Then $\phi^*(x) g_\lambda(x) = \max(0, g_\lambda(x))$ at every x, so $N_\lambda(\phi^*) = \mathbb{E}[\max(0, 1 - \lambda \cdot Z)]$. For $\phi \in \Phi_a$,

$$\begin{aligned}
P(\phi) &= N_\lambda(\phi) + \textstyle\sum_k \lambda_k \mathbb{E}[\phi(X)\,\ell_k(X)] && \text{by Lemma A.2(i)} \\
&\le N_\lambda(\phi^*) + \textstyle\sum_k \lambda_k a_k && \text{by Lemma A.2(ii) and (6)} \\
&= D(\lambda; a).
\end{aligned}$$

(ii) Taking the supremum over $\phi \in \Phi_a$ in (i) gives $P^*(a) \le D(\lambda; a)$ for every $\lambda \ge 0$. For the multiplier λ of the proof of Proposition A.1(b), (9) with this ϕ^* gives $D(\lambda; a) = N_\lambda(\phi^*) + \lambda \cdot a \le P^*(a)$. Hence this λ attains the minimum, which equals $P^*(a)$.

(iii) Part (b3) of the proof of Proposition A.1 uses only (9) for $\phi = \phi^*$, that is, $D(\lambda; a) \le P^*(a)$. Every $\lambda \in \Lambda^*(a)$ satisfies it by (ii). □

Proposition A.2 (Weak monotonicity of prices)**.** *Suppose that every $\ell_k(X)$ is integrable. For $a, a' \in (0, \infty)^K$, $\lambda \in \Lambda^*(a)$ and $\lambda' \in \Lambda^*(a')$,*

$$P^*(a') \le P^*(a) + \lambda \cdot (a' - a) \qquad \text{and} \qquad (\lambda' - \lambda) \cdot (a' - a) \le 0.$$

In particular, loosening level k alone never raises its price. That is, $\lambda'_k \le \lambda_k$ for $a' = a + te_k$ with $t > 0$, where e_k is the k-th unit vector.

The first inequality identifies λ_k as the rate at which the largest share passed increases when level k is relaxed. Proposition A.2 is a standard sensitivity result of convex duality (Boyd and Vandenberghe, 2004, Section 5.6) and requires no assumption on the distribution of X. We include its short proof for completeness.

Proof of Proposition A.2. For the first inequality,

$$\begin{aligned} P^*(a') &= \min_{\lambda'' \ge 0} D(\lambda''; a') \le D(\lambda; a') && \text{by Lemma A.3(ii)} \\ &= D(\lambda; a) + \lambda \cdot (a' - a) && \text{since } D(\lambda; \cdot) \text{ is affine with slope } \lambda \\ &= P^*(a) + \lambda \cdot (a' - a) && \text{since } \lambda \in \Lambda^*(a). \end{aligned}$$

For the second, exchanging the roles of a and a' in the first inequality gives $P^*(a) \le P^*(a') + \lambda' \cdot (a - a')$. Adding the two inequalities,

$$0 \le \lambda \cdot (a' - a) + \lambda' \cdot (a - a') = -(\lambda' - \lambda) \cdot (a' - a).$$

With $a' - a = te_k$ this reads $t(\lambda'_k - \lambda_k) \le 0$. □

A.3 Uniqueness and strict monotonicity

Strict statements require the boundary of the passed set to be non-degenerate.

Assumption A.1. *Z has a bounded continuous density f_Z on a bounded subset of $(0, \infty)^K$. For $\lambda \ne 0$, let σ be the surface measure on the hyperplane $\{z \mid \lambda \cdot z = 1\}$ and $\mu_\lambda = f_Z \, d\sigma / \|\lambda\|$. For every $\lambda \ne 0$, the coordinates $z_1, \dots, z_K$ are linearly independent in $L_2(\mu_\lambda)$.*

For coverage and severity, $Z = (p(X), p(X)c(X))$, and the linear independence fails only if $c(x)$ is constant along the boundary of the passed set.

Notation for the strict results. For $\lambda \ne 0$, let $H_\lambda = \{z \mid \lambda \cdot z = 1\}$, let $M_k(\lambda) = \mathbb{E}[Z_k \mathbf{1}\{\lambda \cdot Z < 1\}]$ be the use of constraint k by the rule that passes $\{x \mid \lambda \cdot Z < 1\}$, and let $\langle u, v \rangle = \int uv \, d\mu_\lambda$. For $J \subseteq \{1, \dots, K\}$, $G_J(\lambda) = (\langle z_i, z_j \rangle)_{i,j \in J}$. Let J be the set of binding constraints, those with a positive price $\lambda^*_k(a)$, and write λ_J and a_J for the subvectors with indices in J. The rows and columns of G_J are indexed by the elements of J.

Lemma A.4 (Derivative along the boundary)**.** *Under Assumption A.1, for $\lambda \neq 0$, every j and every bounded continuous q,*

$$\frac{\partial}{\partial \lambda_j}\mathbb{E}\big[q(Z)\mathbf{1}\{\lambda \cdot Z < 1\}\big] = -\int_{H_\lambda} q(z)\, z_j \, d\mu_\lambda(z).$$

In particular $\partial M_i/\partial \lambda_j = -\langle z_i, z_j\rangle$, and $G_J(\lambda)$ is positive definite for every J.

Proof sketch. A small increase $d\lambda_j$ moves the boundary of the half-space $\{\lambda \cdot z < 1\}$ inward. At a point z of H_λ the normal displacement is $z_j\, d\lambda_j/\|\lambda\|$, and the mass removed is $f_Z(z)$ times this displacement. Integrating over H_λ gives the formula, and boundedness of q, Z and f_Z justifies the exchange of limit and integral. For $v \in \mathbb{R}^J$, $v^\top G_J v = \int (\sum_{j\in J} v_j z_j)^2 \, d\mu_\lambda$, which is positive for $v \neq 0$ by linear independence. □

Proposition A.3 (Uniqueness and strict monotonicity of prices and weights)**.** *Suppose Assumption A.1 holds.*

(a) For every $a \in (0,\infty)^K$, the prices are unique, that is, $\Lambda^*(a) = \{\lambda^*(a)\}$. If $\lambda^*(a) \neq 0$, the weights $w_k(\lambda^*(a))$ are unique as well.

(b) Suppose that at levels a the binding set J is non-empty and the other constraints have slack. Then λ^* is continuously differentiable near a, with

$$\frac{\partial \lambda^*_J}{\partial a_J} = -G_J^{-1}, \qquad \frac{\partial \lambda^*_k}{\partial a_k} = -\big(G_J^{-1}\big)_{kk} < 0 \quad (k \in J).$$

(c) If moreover $|J| \geq 2$ and $(G_J^{-1})_{jk} \leq 0$ for every $j \in J \setminus \{k\}$, then $\partial w_k(\lambda^*(a))/\partial a_k < 0$.

Part (a) holds for any number of constraints. The weights are undefined only when $\lambda^*(a) = 0$. In that case no constraint binds, so every case can be passed and no score is needed. The condition $(G_J^{-1})_{jk} \leq 0$ says that loosening constraint k does not lower the price of any other binding constraint, so the constraints are substitutes. For $K = 2$ it always holds, because $(G^{-1})_{12} = -G_{12}/\det G$ and $G_{12} > 0$ when both losses are positive. For $K \geq 3$ it can fail even though every entry of G_J is positive, as shown below. The price of a constraint thus always responds to its own level, but its weight in the score does so only when the constraints are substitutes. With more than two constraints, the binding set can also change several times as the levels vary, so the levels need not split into the three regimes of Theorem 1.

Proof of Proposition A.3. (a) The function $D(\cdot\,; a)$ is defined on $\mathbb{R}^K$. Its gradient at λ is $a - M(\lambda)$, so by Lemma A.4 its Hessian at every $\lambda \neq 0$ is $G(\lambda) = G_{\{1,\dots,K\}}(\lambda)$, which is positive definite. Suppose that $\lambda, \lambda' \in \Lambda^*(a)$ with $\lambda \neq \lambda'$. Since $\Lambda^*(a)$ is convex, $D(\cdot\,; a)$ is constant on the segment $\lambda_t = \lambda + t(\lambda' - \lambda)$, $t \in [0,1]$. The segment lies in $[0,\infty)^K$. It passes through 0 at most at one endpoint, because $(1-t)\lambda + t\lambda' = 0$ with $t \in (0,1)$ and $\lambda, \lambda' \geq 0$ forces $\lambda = \lambda' = 0$. At every other point,

$$\frac{d^2}{dt^2} D(\lambda_t; a) = (\lambda' - \lambda)^\top G(\lambda_t)\,(\lambda' - \lambda) > 0,$$

which contradicts constancy. Hence $\Lambda^*(a) = \{\lambda^*(a)\}$. If $\lambda^*(a) \neq 0$, then $\sum_j s_j \lambda_j^*(a) > 0$ and the weights $w_k(\lambda^*(a)) = s_k \lambda_k^*(a)/\sum_j s_j \lambda_j^*(a)$ are unique.

(b) Let a^0 be the levels of part (b), let $\lambda^0 = \lambda^*(a^0)$ have binding set J, and suppose that $M_i(\lambda^0) < a_i^0$ for $i \notin J$.

The binding constraints are used in full. The density of Z gives $\Pr(\lambda^0 \cdot Z = 1) = 0$, so by Lemma A.3(iii) and Proposition A.1(b) the optimal rule is $\phi^0(x) = \mathbf{1}\{\lambda^0 \cdot Z < 1\}$ up to a null set, and $M_j(\lambda^0) = a_j^0$ for $j \in J$.

Implicit function theorem. Consider $F(\lambda_J, a_J) = M_J(\lambda_J, 0) - a_J$, with the coordinates outside J set to zero. By Lemma A.4, F is continuously differentiable with $\partial F/\partial \lambda_J = -G_J$, which is invertible. So there are a neighbourhood of a_J^0 and a continuously differentiable map $a_J \mapsto \lambda_J(a_J)$ with $\lambda_J(a_J^0) = \lambda_J^0$ and $M_J(\lambda_J(a_J), 0) = a_J$. Differentiating this identity gives $-G_J\, \partial \lambda_J/\partial a_J = I$.

The solution is a minimizer. Let $\lambda^*(a) = (\lambda_J(a_J), 0)$. By continuity, on a small enough neighbourhood U of a^0 we have $\lambda_j^*(a) > 0$ for $j \in J$ and $M_i(\lambda^*(a)) < a_i$ for $i \notin J$. The rule $\mathbf{1}\{\lambda^*(a) \cdot Z < 1\}$ then has the form (5), belongs to Φ_a, and uses every constraint with a positive multiplier in full. By Proposition A.1(a) it is optimal, and by Lemma A.2(i) and (7), $D(\lambda^*(a); a) = N_\lambda(\phi) + \lambda \cdot a = P(\phi) = P^*(a)$, so $\lambda^*(a) \in \Lambda^*(a)$. By part (a), it is the unique element.

Sign. The matrix G_J^{-1} is positive definite, so its diagonal entries $e_k^\top G_J^{-1} e_k$ are positive.

(c) Let $c = \partial \lambda^*/\partial a_k$. By the first part and the condition $(G_J^{-1})_{jk} \le 0$,

$$c_k = -\big(G_J^{-1}\big)_{kk} < 0, \qquad c_j = -\big(G_J^{-1}\big)_{jk} \ge 0 \;\; (j \in J \setminus \{k\}), \qquad c_i = 0 \;\; (i \notin J).$$

With $S = \sum_j s_j \lambda_j^* > 0$, the quotient rule gives

$$\frac{\partial w_k}{\partial a_k} = \frac{s_k}{S^2}\Big(c_k S - \lambda_k^* \textstyle\sum_j s_j c_j\Big) = \frac{s_k}{S^2}\Big(c_k \sum_{j \ne k} s_j \lambda_j^* \;-\; \lambda_k^* \sum_{j \ne k} s_j c_j\Big).$$

The first term is negative, because $c_k < 0$ and $\sum_{j \ne k} s_j \lambda_j^* > 0$ as J contains an index other than k. The second term is at most zero, because $\lambda_k^* > 0$ and $c_j \ge 0$. □

The substitutes condition can fail for $K \ge 3$. The matrix

$$G_J = \begin{pmatrix} 1 & 0.8 & 0.8 \\ 0.8 & 1 & 0.5 \\ 0.8 & 0.5 & 1 \end{pmatrix}$$

has positive entries and eigenvalues 0.09, 0.5 and 2.41, so it is positive definite, but $(G_J^{-1})_{23} = 1.27 > 0$. Loosening constraint 3 then lowers the price of constraint 2, and the weight of a constraint may fall when its level tightens.

B Proofs of the main results

B.1 Proof of Theorem 1

We use the notation of Appendix A with $K = 2$, $Z = (p(X), p(X)c(X))$ and $a = (\alpha_{\text{cov}}\pi_1, \alpha_{\text{sev}})$.

Proof of Theorem 1. (a) Form of the optimal rule. By Proposition A.1(b), an optimal rule exists, and there are prices $\lambda \ge 0$ such that every optimal rule passes x when $g_\lambda(x) > 0$ and reviews it

when $g_\lambda(x) < 0$, for almost every x with $g_\lambda(x) \neq 0$. If $\lambda = 0$, then $g_\lambda \equiv 1$ and every optimal rule passes almost every case, so $\Pr(\text{pass} \mid Y = 1) = 1 > \alpha_{\text{cov}}$, a contradiction. Hence $\lambda \neq 0$. Let $S = \lambda_1 + \lambda_2 B > 0$, $\theta^* = \lambda_2 B / S$ and $t^* = 1/S$. Then $g_\lambda(x) = 1 - S\, h_{\theta^*}(x)$, so $g_\lambda(x) > 0$ exactly when $h_{\theta^*}(x) < t^*$. Under Assumption A.1, Z has a density, so $\Pr(g_\lambda(X) = 0) = 0$. This proves (a).

(b) Uniqueness and continuity. Write $\overline{a} = S_{\text{cc}}$ and $G = G_{\{1,2\}}$. The prices $\lambda^*(a)$ are unique by Proposition A.3(a), and $\lambda^*(a) \neq 0$ by (a), so θ^* is unique. Since $D(\cdot; a)$ is continuous in (λ, a) and grows linearly in λ, the maximum theorem of Berge shows that $\lambda^*(a)$ is continuous in a.

Regimes. The coverage-only rule is feasible at a_2 if and only if $a_2 \geq \overline{a}$, and for such a_2 it is optimal with $\lambda_2 = 0$, so $\theta^* = 0$. For the severity-only rule at level a_2, a cut-off on $p(x)c(x)$, the cut-off and hence its coverage use $u(a_2)$ are non-decreasing in a_2. Let $\underline{a} = \sup\{a_2 \mid u(a_2) \leq a_1\}$, as in Section 3. As $a_2 \downarrow 0$, the cut-off t of the severity-only rule tends to 0, so $u(a_2) = \mathbb{E}[p(X)\mathbf{1}\{p(X)c(X) \leq t\}] \to 0$ by dominated convergence, since $p(X)c(X) > 0$ almost surely by Assumption A.1. Since $a_1 > 0$, the set is non-empty and $\underline{a} > 0$. For $a_2 \leq \underline{a}$ the severity-only rule is feasible and optimal with $\lambda_1 = 0$, so $\theta^* = 1$.

Order of the thresholds. At $a_2 = \overline{a}$, the coverage-only rule meets the severity constraint. It is therefore feasible for the problem with the severity constraint alone. Under Assumption A.1, $c(x)$ is not constant on the boundary of the set passed by the coverage-only rule. The two rules thus differ on a set of positive probability. The severity-only rule is the unique rule that passes the largest share under the severity constraint alone, so it passes a strictly larger share than the coverage-only rule. The coverage-only rule passes the largest share under the coverage constraint alone. Hence the severity-only rule violates the coverage constraint at $\overline{a}$, that is, $u(\overline{a}) > a_1$. Since u is non-decreasing and, under Assumption A.1, continuous, $\underline{a} < \overline{a}$.

Both prices are positive in the middle. Let $a_2 \in (\underline{a}, \overline{a})$. If $\lambda_1 = 0$, then $\lambda_2 > 0$ and the optimal rule is a cut-off on $p(x)c(x)$ that uses the severity level in full, which is the severity-only rule. It violates coverage because $a_2 > \underline{a}$. If $\lambda_2 = 0$, the optimal rule is a cut-off on $p(x)$ that uses the coverage level in full, which violates severity because $a_2 < \overline{a}$. Hence $J = \{1, 2\}$, and Proposition A.3 applies with no index outside J.

Sign of the cross effect. Since Z has positive coordinates and $\mu_\lambda \neq 0$, $\langle z_1, z_2 \rangle > 0$. For a 2×2 matrix,

$$\left(G^{-1}\right)_{12} = -\frac{\langle z_1, z_2 \rangle}{\det G} < 0,$$

so the substitutes condition holds with $k = 2$. Proposition A.3 with $s = (1, B)$ gives $\partial\theta^* / \partial a_2 < 0$ on the middle regime.

Continuity at the ends. By continuity of $\lambda^*(a)$, $\lambda_2^* \to 0$ as $a_2 \uparrow \overline{a}$ and $\lambda_1^* \to 0$ as $a_2 \downarrow \underline{a}$, so θ^* joins the three regimes continuously. □

B.2 Guarantees of the calibrated rule

Lemma B.1 (Guarantees)**.** *If the calibration cases and the test case are i.i.d., and $\hat{p}$, $\hat{c}$ and θ do not depend on them, the calibrated rule satisfies (1) and (2).*

Proof. Conformal risk control is stated for a parameter at which the loss does not increase. Applied to the parameter $-t$, its infimum becomes the supremum in (4). At $t = -\infty$ no case passes and every loss below is zero.

Severity. The losses $C_i\,\mathbf{1}\{Y_i = 1,\ \hat{h}_\theta(X_i) < t\} \in [0, B]$ of the calibration cases and of the test case are i.i.d. and non-decreasing in t, so (4) gives $\mathbb{E}[C\,\mathbf{1}\{Y = 1,\ \hat{h}_\theta(X) < \hat{t}_{\text{sev}}\}] \le \alpha_{\text{sev}}$.

Coverage. Given the labels of the calibration cases and $Y = 1$ for the test case, the n_1 adverse calibration cases and the test case are i.i.d. draws of X given $Y = 1$. The losses $\mathbf{1}\{\hat{h}_\theta(X_i) < t\}$ are non-decreasing in t, so (4) gives $\Pr(\hat{h}_\theta(X) < \hat{t}_{\text{cov}} \mid Y = 1) \le \alpha_{\text{cov}}$.

Both. Since $\hat{t} \le \hat{t}_{\text{cov}}$ and $\hat{t} \le \hat{t}_{\text{sev}}$, the passed set $\{\hat{h}_\theta(X) < \hat{t}\}$ is contained in each of $\{\hat{h}_\theta(X) < \hat{t}_{\text{cov}}\}$ and $\{\hat{h}_\theta(X) < \hat{t}_{\text{sev}}\}$. Both losses are non-negative, so each risk of the calibrated rule is at most the corresponding bound above. □

B.3 Proof of Theorem 2(a) and (b)

Lemma B.2 (Smoothness of the dual function)**.** *Let $K = 2$, fix a and write $D(\lambda) = D(\lambda; a)$. Under Assumption A.1, D is twice continuously differentiable on $[0, \infty)^2 \setminus \{0\}$, with*

$$\nabla D(\lambda) = a - \big(M_1(\lambda), M_2(\lambda)\big), \qquad \nabla^2 D(\lambda) = G(\lambda),$$

which is positive definite. The optimal multiplier λ^* is its unique minimizer.

Proof. The derivative of $\max(0, 1-\lambda\cdot z)$ in λ is $-z\mathbf{1}\{\lambda\cdot z < 1\}$ except on H_λ, which has probability zero. Since Z is bounded, differentiation under the expectation gives the gradient, and Lemma A.4 gives the Hessian. Lemma A.3(ii) and Proposition A.3(a) give the last claim. □

Proof of Theorem 2(a). Let $\hat{Z} = (\hat{p}(X), \hat{p}(X)\hat{c}(X))$ and $\hat{Z}_i$ its value at x_i. The dual of the plug-in linear program, divided by n_U, is

$$\hat{D}(\lambda) = \frac{1}{n_U}\sum_{i=1}^{n_U}\Big(\max(0,\ 1 - \lambda\cdot\hat{Z}_i) + \lambda_1\alpha_{\text{cov}}\,\hat{p}(x_i) + \lambda_2\alpha_{\text{sev}}\Big),$$

and its minimizer is $\hat{\lambda}$. It is an average of independent convex functions of λ. Its population counterpart is

$$\tilde{D}(\lambda) = \mathbb{E}\big[\max(0,\ 1 - \lambda\cdot\hat{Z})\big] + \lambda\cdot\tilde{a}, \qquad \tilde{a} = \big(\alpha_{\text{cov}}\mathbb{E}[\hat{p}(X)],\ \alpha_{\text{sev}}\big),$$

the dual function of the problem built on $\hat{p}$ and $\hat{c}$. Its minimizer $\tilde{\lambda}$ gives $\tilde{\theta}$.

Lemma B.2 applies to $\hat{Z}$ because part (a) assumes Assumption A.1 for $\hat{Z}$. Hence $\tilde{D}$ is twice continuously differentiable, its Hessian $\tilde{G} = \tilde{G}(\tilde{\lambda})$ is positive definite, and $\tilde{\lambda}$ is its unique minimizer. Both constraints bind in this problem, so $\tilde{\lambda}$ lies in the interior of $[0, \infty)^2$. Each summand of $\hat{D}$ is differentiable at $\tilde{\lambda}$ with probability one, with derivative

$$\tilde{\xi}(X) = \big(\alpha_{\text{cov}}\,\hat{p}(X),\ \alpha_{\text{sev}}\big) - \hat{Z}\,\mathbf{1}\{\tilde{\lambda}\cdot\hat{Z} < 1\}.$$

This derivative is bounded because $\hat{Z}$ is bounded. These are the conditions of the basic corollary of Hjort and Pollard (1993) for minimizers of convex processes. It gives $\sqrt{n_U}(\hat{\lambda} - \tilde{\lambda}) \to N(0, \tilde{G}^{-1}\tilde{\Sigma}\tilde{G}^{-1})$ in distribution, where $\tilde{\Sigma}$ is the covariance of $\tilde{\xi}(X)$. The map $\lambda \mapsto \lambda_2 B/(\lambda_1 + \lambda_2 B)$ is smooth near $\tilde{\lambda}$, and the delta method gives the asymptotic normality of $\sqrt{n_U}(\hat{\theta} - \tilde{\theta})$.

With exact estimates, $\hat{Z} = Z$, $\tilde{D} = D$ and $\tilde{\lambda} = \lambda^*$, so $\tilde{\theta} = \theta^*$. □

Proof of Theorem 2(b). Let $\tilde{D}$ be the dual function of the problem built on $\hat{p}$ and $\hat{c}$, as in the proof of (a), and let $\tilde{\lambda}$ minimize it over $[0,\infty)^2$. By Lemma B.2, there are $\omega > 0$ and $s_0 \in (0, \min(\lambda_1^*, \lambda_2^*)]$ such that D is ω-strongly convex on the ball $U = \{\lambda \mid \|\lambda - \lambda^*\| \le s_0\}$. This ball lies in $(0,\infty)^2$. Let e be as in Section 4 and $\varepsilon = (\|\lambda^*\|_\infty + s_0)\, e$.

Step 1. The two dual functions are close on U. For $\lambda \in U$, $\|\lambda\|_\infty \le \|\lambda^*\|_\infty + s_0$. Since $u \mapsto \max(0, 1-u)$ is 1-Lipschitz,

$$\big|\max(0, 1 - \lambda \cdot \hat{Z}) - \max(0, 1 - \lambda \cdot Z)\big| \le \|\lambda\|_\infty \big(|\hat{p}(X) - p(X)| + |\hat{p}(X)\hat{c}(X) - p(X)c(X)|\big),$$

and the linear terms differ by $\lambda_1 \alpha_{\mathrm{cov}} |\mathbb{E}[\hat{p}(X) - p(X)]|$. Taking expectations, $|\tilde{D}(\lambda) - D(\lambda)| \le \varepsilon$ on U.

Step 2. The minimizer moves little. Both constraints bind, so $\nabla D(\lambda^*) = 0$. For λ with $\|\lambda - \lambda^*\| = s \le s_0$, strong convexity and Step 1 give

$$\tilde{D}(\lambda) \ge D(\lambda) - \varepsilon \ge D(\lambda^*) + \tfrac{\omega}{2} s^2 - \varepsilon \ge \tilde{D}(\lambda^*) + \tfrac{\omega}{2} s^2 - 2\varepsilon.$$

If $s > 2\sqrt{\varepsilon/\omega}$, the right side exceeds $\tilde{D}(\lambda^*)$. A convex function that is larger on a sphere than at its centre attains its minimum inside the sphere. Hence, whenever $2\sqrt{\varepsilon/\omega} < s_0$,

$$\|\tilde{\lambda} - \lambda^*\| \le 2\sqrt{\varepsilon/\omega}.$$

Step 3. From prices to weights. Let L_θ be the Lipschitz constant of $\lambda \mapsto \lambda_2 B/(\lambda_1 + \lambda_2 B)$ on U. Then

$$|\tilde{\theta} - \theta^*| \le L_\theta \|\tilde{\lambda} - \lambda^*\| \le C_0 \sqrt{e}, \qquad C_0 = 2L_\theta \sqrt{\frac{\|\lambda^*\|_\infty + s_0}{\omega}}.$$

The condition $2\sqrt{\varepsilon/\omega} < s_0$ reads $e < \omega s_0^2 / (4(\|\lambda^*\|_\infty + s_0))$. This is the meaning of a small enough e in Theorem 2(b). □

B.4 Proof of Theorem 2(c)

Notation. For a score s and a cut-off t, the rule passes $\{s(X) < t\}$. Its risks are $R_{\mathrm{cov}}(s,t) = \Pr(s(X) < t \mid Y = 1)$ and $R_{\mathrm{sev}}(s,t) = \mathbb{E}[C\,\mathbf{1}\{Y = 1,\ s(X) < t\}]$, and its share sent to review is $\mathrm{Rev}(s,t) = \Pr(s(X) \ge t)$. For levels $b = (b_{\mathrm{cov}}, b_{\mathrm{sev}})$, let $\tau_s(b) = \sup\{t \mid R_{\mathrm{cov}}(s,t) \le b_{\mathrm{cov}},\ R_{\mathrm{sev}}(s,t) \le b_{\mathrm{sev}}\}$ and $V(s,b) = \mathrm{Rev}(s, \tau_s(b))$, the smallest share sent to review by a cut-off on s. Write $\alpha = (\alpha_{\mathrm{cov}}, \alpha_{\mathrm{sev}})$. By Proposition A.1, $R^* = V(h_{\theta^*}, \alpha)$ is the smallest share sent to review among all rules that satisfy both constraints.

Assumption B.1. *For θ in a neighbourhood Θ_0 of θ^*,*

(i) outside an event E of probability at most ρ, $|\hat{p}(X) - p(X)| \le \eta$ and $|\hat{p}(X)\hat{c}(X) - p(X)c(X)|/B \le \eta$,

(ii) $h_\theta(X)$ has a density bounded by f, and $\hat{h}_\theta(X)$ has a continuous distribution,

(iii) $V(h_\theta, \alpha - d) \le V(h_\theta, \alpha) + L_{\mathrm{cov}} d_1 + L_{\mathrm{sev}} d_2$ for small $d \ge 0$.

Condition (i) allows a small set of cases on which the estimates are poor. Condition (iii) bounds how fast the share sent to review grows as the levels tighten, and a constraint with slack contributes

nothing to it. The dependence on the weight needs no assumption. Lemma B.3 below derives it from Assumption A.1.

The share sent to review near the optimal weight. For θ near θ^*, a cut-off on h_θ must stop as soon as one of the two levels is used in full. A weight below or above θ^* exhausts one level before the other, so the share sent to review has a kink at θ^* rather than a smooth minimum.

Lemma B.3 (Kink at the optimal weight)**.** *Suppose Assumption A.1 holds and both constraints bind. There are constants $H_+, H_- > 0$ such that*

$$V(h_\theta, \alpha) = R^* + H_+(\theta - \theta^*) + o(\theta - \theta^*) \ \ (\theta > \theta^*), \qquad V(h_\theta, \alpha) = R^* + H_-(\theta^* - \theta) + o(\theta^* - \theta) \ \ (\theta < \theta^*).$$

On one side of θ^* the coverage level is used in full and the severity level has slack. On the other side the roles are reversed. The slope on each side is proportional to the price of the level left unused. In particular, for every $H > \max(H_+, H_-)$ there is a neighbourhood of θ^* on which $V(h_\theta, \alpha) \le R^* + H|\theta - \theta^*|$.

Proof. We use the notation of Appendix A.3 with $Z = (p(X), p(X)c(X))$ and $a = (\alpha_{\mathrm{cov}}\pi_1, \alpha_{\mathrm{sev}})$. Let $P(\lambda) = \Pr(\lambda \cdot Z < 1)$ be the share passed by the rule with prices λ, let $G = G(\lambda^*)$ and $m = G\lambda^*$.

Step 1. Derivatives at λ^*. By Lemma A.4 with $q = 1$, $\partial P/\partial \lambda_j = -\int z_j \, d\mu_\lambda$. On the hyperplane H_λ we have $\lambda \cdot z = 1$, so $\int z_j \, d\mu_\lambda = \int z_j \, (\lambda \cdot z) \, d\mu_\lambda = (G(\lambda)\lambda)_j$. Hence $\nabla P(\lambda^*) = -m$. Both coordinates of Z are positive and $\mu_{\lambda^*} \neq 0$, since G is positive definite. Hence $m > 0$. By Lemma A.4, $\nabla M_k(\lambda^*) = -Ge_k$.

Step 2. Cut-offs that exhaust one level. The cut-off rule $\{h_\theta < t\}$ is the rule with prices $\lambda = s\, r(\theta)$, where $r(\theta) = (1-\theta, \ \theta/B)$ and $s = 1/t$. A larger s passes fewer cases. At θ^* we have $\lambda^* = s^* r(\theta^*)$ for some $s^* > 0$. Since $\partial M_k(s\, r(\theta^*))/\partial s = -(G\lambda^*)_k/s^* = -m_k/s^* < 0$ at $s = s^*$, the implicit function theorem gives continuously differentiable maps $s_1(\theta)$ and $s_2(\theta)$ near θ^* with $M_k(s_k(\theta) r(\theta)) = a_k$ and $s_k(\theta^*) = s^*$. The rule that uses the level of constraint k in full passes the share $P_k(\theta) = P(s_k(\theta) r(\theta))$. The best cut-off on h_θ is the larger of the two values of s, so

$$V(h_\theta, \alpha) = 1 - \min\big(P_1(\theta), \, P_2(\theta)\big), \qquad P_1(\theta^*) = P_2(\theta^*) = 1 - R^*.$$

Step 3. Slopes. Let $\lambda_k(\theta) = s_k(\theta) r(\theta)$. This curve lies on the level set $\{M_k = a_k\}$, so its tangent is orthogonal to Ge_k. Hence $\lambda_1'(\theta^*)$ is parallel to $G^{-1}e_2$ and $\lambda_2'(\theta^*)$ is parallel to $G^{-1}e_1$. The weight of λ is $\theta(\lambda) = \lambda_2 B/(\lambda_1 + \lambda_2 B)$, with gradient $\kappa_0 w$ at λ^*, where

$$w = (-\lambda_2^*, \ \lambda_1^*), \qquad \kappa_0 = \frac{B}{(\lambda_1^* + B\lambda_2^*)^2}.$$

Let $y = G^{-1}w$. Along the first curve $\theta(\lambda_1(\theta)) = \theta$, so $\lambda_1'(\theta^*) = G^{-1}e_2/(\kappa_0 \, w \cdot G^{-1}e_2) = G^{-1}e_2/(\kappa_0 y_2)$, by the symmetry of G. Likewise $\lambda_2'(\theta^*) = G^{-1}e_1/(\kappa_0 y_1)$. By Step 1,

$$P_1'(\theta^*) = -\frac{m \cdot G^{-1}e_2}{\kappa_0 y_2} = -\frac{\lambda_2^*}{\kappa_0 y_2}, \qquad P_2'(\theta^*) = -\frac{\lambda_1^*}{\kappa_0 y_1}.$$

These require $y_1 \neq 0$ and $y_2 \neq 0$, which Step 4 shows.

Step 4. Opposite signs. We have $m \cdot y = \lambda^{*\top} G G^{-1} w = \lambda^* \cdot w = 0$. Since $m > 0$ and $y \neq 0$, the coordinates y_1 and y_2 are non-zero and of opposite signs. Both prices are positive, so $P_1'(\theta^*)$ and $P_2'(\theta^*)$ are non-zero and of opposite signs.

Step 5. The kink. For $\theta > \theta^*$, the minimum in Step 2 is attained by the rule whose derivative is negative, so $V(h_\theta, \alpha)$ has right derivative $H_+ = -\min(P_1'(\theta^*), P_2'(\theta^*)) > 0$. For $\theta < \theta^*$ it is attained by the other rule, and the left derivative is $-H_-$ with $H_- = \max(P_1'(\theta^*), P_2'(\theta^*)) > 0$. If $y_2 > 0$, the coverage level is used in full to the right of θ^* and $H_+ = \lambda_2^*/(\kappa_0 y_2)$ is proportional to the price of severity, the level left unused. The case $y_2 < 0$ is symmetric. The last claim follows from the definition of the derivative. □

The kink concerns the share sent to review as a function of θ only. Theorem 1 and Theorem 2(a) and (b) work with the dual function, which is smooth in λ, and are not affected.

Procedure. The estimates $\hat{p}(x)$ and $\hat{c}(x)$ are fitted on separate data. The weight $\hat{\theta}$ is the plug-in dual estimate of Section 4.2. The calibration sample has n cases, of which n_1 are defaults. With $\hat{R}_{\mathrm{cov}}(t)$ the share of calibration defaults with $\hat{h}_{\hat{\theta}}(X_i) < t$ and $\hat{R}_{\mathrm{sev}}(t) = n^{-1}\sum_i C_i \mathbf{1}\{Y_i = 1,\ \hat{h}_{\hat{\theta}}(X_i) < t\}$, the multi-risk CRC uses

$$\hat{t} = \min(\hat{t}_{\mathrm{cov}}, \hat{t}_{\mathrm{sev}}), \quad \hat{t}_{\mathrm{cov}} = \sup\Big\{t \;\Big|\; \tfrac{n_1 \hat{R}_{\mathrm{cov}}(t)+1}{n_1+1} \le \alpha_{\mathrm{cov}}\Big\}, \quad \hat{t}_{\mathrm{sev}} = \sup\Big\{t \;\Big|\; \tfrac{n \hat{R}_{\mathrm{sev}}(t)+B}{n+1} \le \alpha_{\mathrm{sev}}\Big\}.$$

Under Assumption B.1(i), outside the exceptional event E, $\hat{h}_\theta(X) - h_\theta(X) = (1-\theta)(\hat{p}(X) - p(X)) + \theta(\hat{p}(X)\hat{c}(X) - p(X)c(X))/B$ is a convex combination of two errors bounded by η, so $|\hat{h}_\theta(X) - h_\theta(X)| \le \eta$ for every θ at once.

Lemma B.4 (Concentration of the calibration risks)**.** *Fix a score s that does not depend on the calibration sample and let $\delta \in (0,1)$. Conditionally on n_1, with probability at least $1-\delta$,*

$$\sup_t |\hat{R}_{\mathrm{cov}}(t) - R_{\mathrm{cov}}(s,t)| \le \epsilon_{\mathrm{cov}} = \sqrt{\frac{\log(4/\delta)}{2n_1}},$$

$$\sup_t |\hat{R}_{\mathrm{sev}}(t) - R_{\mathrm{sev}}(s,t)| \le \epsilon_{\mathrm{sev}} = B\Big(\frac{1}{J} + \sqrt{\frac{\log(4(J+1)/\delta)}{2n}}\Big),$$

with $J = \lceil \sqrt{n} \rceil$.

Proof. Given n_1, the scores of the calibration defaults are independent draws from the law of $s(X)$ given $Y = 1$, and $\hat{R}_{\mathrm{cov}}$ is their empirical distribution function. The DKW inequality with the constant of Massart (1990) gives the first bound with probability at least $1 - \delta/2$. For the second, $R_{\mathrm{sev}}(s,\cdot)$ is continuous and non-decreasing with range in $[0, B]$, so there are cut-offs $t_0 < \cdots < t_J$, with $t_0 = -\infty$ and $t_J = +\infty$, between which it rises by at most B/J. The Hoeffding inequality at these $J+1$ points and a union bound give $|\hat{R}_{\mathrm{sev}}(t_j) - R_{\mathrm{sev}}(s,t_j)| \le B\sqrt{\log(4(J+1)/\delta)/(2n)}$ for all j with probability at least $1 - \delta/2$. For $t \in [t_{j-1}, t_j]$, monotonicity gives $\hat{R}_{\mathrm{sev}}(t) \le \hat{R}_{\mathrm{sev}}(t_j) \le R_{\mathrm{sev}}(s,t) + \epsilon_{\mathrm{sev}}$, and the lower bound follows in the same way. □

Lemma B.5 (Position of the calibrated cut-off)**.** *On the event of Lemma B.4, $\hat{t} \ge \tau_s(\alpha - \gamma)$ with $\gamma = (\gamma_{\mathrm{cov}}, \gamma_{\mathrm{sev}}) = (\epsilon_{\mathrm{cov}} + 1/n_1,\ \epsilon_{\mathrm{sev}} + B/n)$. Hence the share sent to review by the calibrated rule is at most $V(s, \alpha - \gamma)$.*

Proof. If $R_{\mathrm{cov}}(s,t) \le \alpha_{\mathrm{cov}} - \gamma_{\mathrm{cov}}$, then $\hat{R}_{\mathrm{cov}}(t) \le \alpha_{\mathrm{cov}} - 1/n_1$ and $n_1 \hat{R}_{\mathrm{cov}}(t) + 1 \le n_1 \alpha_{\mathrm{cov}} \le (n_1+1)\alpha_{\mathrm{cov}}$, so $t \le \hat{t}_{\mathrm{cov}}$. The same argument with B in place of 1 gives $t \le \hat{t}_{\mathrm{sev}}$. Taking the supremum over such t gives the first claim, and $\mathrm{Rev}(s,\cdot)$ is non-increasing. □

Lemma B.6 (From estimated to true scores)**.** *Under Assumption B.1(i)–(ii), for $\theta \in \Theta_0$ and levels b with $b > \beta$,*

$$V(\hat{h}_\theta, b) \le V(h_\theta, b-\beta) + 2f\eta + \rho, \qquad \beta = (\rho/\pi_1,\ B\rho).$$

Proof. Let $t < \tau_{h_\theta}(b-\beta)$ and consider the rule that passes $\{x \mid \hat{h}_\theta(x) < t-\eta\}$. Outside E this set lies inside $\{x \mid h_\theta(x) < t\}$, so the passed set lies inside $\{x \mid h_\theta(x) < t\} \cup E$. Since the losses are non-negative, its coverage risk is at most $R_{\mathrm{cov}}(h_\theta, t) + \Pr(E \mid Y = 1) \le b_{\mathrm{cov}} - \rho/\pi_1 + \rho/\pi_1$, and its severity risk is at most $R_{\mathrm{sev}}(h_\theta, t) + B\Pr(E) \le b_{\mathrm{sev}}$. The rule is feasible at levels b, so $V(\hat{h}_\theta, b) \le \Pr(\hat{h}_\theta(X) \ge t-\eta)$. Outside E, $h_\theta(X) < t - 2\eta$ implies $\hat{h}_\theta(X) < t-\eta$, hence $\Pr(\hat{h}_\theta(X) \ge t-\eta) \le \Pr(h_\theta(X) \ge t-2\eta) + \rho \le \mathrm{Rev}(h_\theta, t) + 2f\eta + \rho$. Letting $t \uparrow \tau_{h_\theta}(b-\beta)$ completes the proof. □

Theorem B.1 (Precise form of Theorem 2(c))**.** *Suppose Assumptions A.1 and B.1 hold, both constraints bind, and $\hat{\theta} \in \Theta_0$, where Θ_0 also lies in the neighbourhood of Lemma B.3 for a constant H. Suppose also that $\gamma + \beta$, defined below and in Lemma B.6, is small enough that $\alpha - \gamma - \beta > 0$ and Assumption B.1(iii) applies with $d = \gamma + \beta$, which holds for large n_1 and n and small ρ. With probability at least $1-\delta$ over the calibration sample, given $\hat{\theta}$ and n_1, the calibrated rule with $\hat{h}_{\hat{\theta}}$ sends to review a share of new cases of at most*

$$R^* + H|\hat{\theta} - \theta^*| + L_{\mathrm{cov}}\Big(\gamma_{\mathrm{cov}} + \frac{\rho}{\pi_1}\Big) + L_{\mathrm{sev}}(\gamma_{\mathrm{sev}} + B\rho) + 2f\eta + \rho,$$

with $\gamma_{\mathrm{cov}} = \epsilon_{\mathrm{cov}} + 1/n_1$ and $\gamma_{\mathrm{sev}} = \epsilon_{\mathrm{sev}} + B/n$ as in Lemma B.4.

Proof of Theorem B.1. The score $\hat{h}_{\hat{\theta}}$ does not depend on the calibration sample. By Lemma B.5, then Lemma B.6 with $b = \alpha - \gamma$, then Assumption B.1(iii) and Lemma B.3,

$$\mathrm{Rev}(\hat{h}_{\hat{\theta}}, \hat{t}) \le V(\hat{h}_{\hat{\theta}}, \alpha-\gamma) \le V(h_{\hat{\theta}}, \alpha-\gamma-\beta) + 2f\eta + \rho \le R^* + H|\hat{\theta} - \theta^*| + L\cdot(\gamma+\beta) + 2f\eta + \rho. \ \square$$

From the precise form to Theorem 2(c). Since $\log(4/\delta) > 1$ and $1/n_1 \le 1/\sqrt{n_1}$, $\gamma_{\mathrm{cov}} \le 2\sqrt{\log(4/\delta)/n_1}$. With $J = \lceil \sqrt{n} \rceil$, both B/J and B/n are at most $B/\sqrt{n}$. For $\delta \le 1/2$ and $n \ge 2$, the logarithms are therefore bounded by a multiple of $\log(1/\delta)$ and $\log(n/\delta)$, so $\gamma_{\mathrm{cov}} \le c\sqrt{\log(1/\delta)/n_1}$ and $\gamma_{\mathrm{sev}} \le cB\sqrt{\log(n/\delta)/n}$ for an absolute constant c. The remaining terms are at most $C_2(\eta + \rho)$ with $C_2 = \max(2f,\ L_{\mathrm{cov}}/\pi_1 + BL_{\mathrm{sev}} + 1)$. Theorem 2(c) follows with $C_1 = c\max(L_{\mathrm{cov}}, L_{\mathrm{sev}})$.

C Details of the simulation study

Data-generating process. The features $X = (X_1, X_2, X_3)$ are independent standard normal variables, $B = 1$, and $\sigma(z) = 1/(1+e^{-z})$. The default probability and the expected loss given default are

$$p(x) = \sigma\big(-2.2 + x_1 + 0.5\,x_1x_2\big), \qquad c(x) = \sigma\big(-1 + 1.5\,x_2 - 0.3\,x_1 + 0.5\,x_3\big).$$

Given $X = x$ and $Y = 1$, the loss is $C \sim \mathrm{Beta}\big(4c(x),\ 4(1-c(x))\big)$, so that $\mathbb{E}[C \mid X = x, Y = 1] = c(x)$. The share of adverse cases is 13.8%, and the mean loss of a default is 0.34. The negative coefficient of x_1 in $c(x)$ places larger losses at a lower default probability, as in the LendingClub loan dataset. All figures use $\alpha_{\mathrm{cov}} = 0.10$. The coverage-only rule leaves an uncovered severity of $S_{\mathrm{cc}} = 0.00528$, which is 11.1% of the expected default loss, and $\underline{a} = 0.00059$.

The optimal rule. With the true p and c, the cut-off on h_θ for a fixed θ passes the cases with the lowest scores for as long as both constraints hold. The best weight passes the largest share. It is found on a grid of θ with step 0.02 and refined twice around the best value with steps 0.001 and 0.0001. On a finite sample, the best cut-off rule solves the plug-in linear program of Section 4.2 up to the cases on its boundary, so this weight agrees with the weight given by the dual variables up to the step of the grid.

Figure 1. Both panels use 400,000 draws. Panel (a) sets $\alpha_{\text{sev}} = 0.5\, S_{\text{cc}}$ and uses a grid of θ with step 0.01. The two separate cut-offs are set one per constraint, the cut-off on p at the coverage level and the cut-off on pc at the severity level, and a case passes only below both. Panel (b) uses 80 levels spaced evenly on a log scale from $0.5\,\underline{a}$ to $1.5\, S_{\text{cc}}$ and a grid of θ with step 0.001. At each level, it records the weights whose rule sends to review at most one case more than the best rule.

Figure 2. The levels are $\alpha_{\text{cov}} = 0.10$ and $\alpha_{\text{sev}} = 0.5\, S_{\text{cc}}$. The reference weight θ^* and the optimal share R^* are computed on 4,000,000 draws. In panel (a), the weight is estimated from the true p and c of $n_U \in \{1{,}000,\ 4{,}000,\ 16{,}000,\ 64{,}000,\ 256{,}000\}$ fresh draws, with 100 repetitions per size. In panel (b), the training, unlabeled and calibration samples each have $n \in \{2{,}000,\ 5{,}000,\ 20{,}000,\ 50{,}000,\ 200{,}000\}$ cases, with 20 repetitions per size. The estimate $\hat{p}$ is a histogram gradient boosting classifier, and $\hat{c}$ a histogram gradient boosting regressor fitted on the defaults only and clipped to $[0, B]$. Both use 200 iterations and a learning rate of 0.1. The share sent to review, the share of defaults left uncovered and the uncovered severity are evaluated on 1,000,000 fresh draws.

Bias from the estimates. The distorted estimates scale c by a factor m and clip it to $[0, B]$, or scale the logit of p by a factor m, with $m \in \{0.80, 0.85, 0.90, 0.95, 0.98, 1.02, 1.05, 1.10, 1.15, 1.20\}$. The weight $\tilde{\theta}$ of each distortion and the weight θ^* are computed on a common draw of 1,000,000 cases, so that sampling noise cancels in their difference.

D Details of the LendingClub experiments

Loans. We use the file of accepted loans from 2007 to the fourth quarter of 2018. We keep loans with a term of 36 months, issued from January 2012 to December 2015, whose status is fully paid, charged off or default. Loans marked as not meeting the credit policy are kept under the same statuses. Every kept loan has matured by the end of 2018, so loans still running at that date do not bias the sample. The cut at 2012 drops earlier years in which some credit features were not yet collected. The sample has 589,488 loans. A loan is a default ($Y = 1$) when it is charged off or in default.

Loss. For a default, the loss is the loan amount minus the total amount paid back, where the payments include recoveries after charge-off and exclude the fee paid to the collection agency. Negative values are set to zero. The loss is thus at most the loan amount, and $B = \$35{,}000$ is the largest loan amount in the sample.

Features. The models use 29 features known at origination. Six describe the loan, namely the amount, the interest rate, the instalment, the grade, the sub-grade and the purpose. Six describe the application, namely the length of employment, the home ownership, the annual income, the verification status of the income, the state and the ratio of debt to income. The other seventeen describe the credit history, such as the lower bound of the FICO range, the age of the oldest credit line, past delinquencies and public records, recent inquiries, open and total accounts, and the balance and utilization of revolving credit. The grade, the sub-grade, the purpose, the home

ownership, the verification status and the state enter as categorical features. No feature records the outcome of the loan.

Samples. The 306,462 loans issued from 2012 to 2014 are split at random into a training sample of 275,816 loans and a validation sample of 30,646 loans. The 283,026 loans issued in 2015 form the pool for calibration and testing.

Models. Both fits use LightGBM with a learning rate of 0.05, 31 leaves per tree, at least 100 loans per leaf and at most 2,000 trees. Training stops when the loss on the validation sample has not improved for 100 trees. The classifier for $\hat{p}$ uses all training loans and the logistic loss. The regressor for the loss rate, the loss divided by the loan amount, uses the training defaults only and the squared loss, so that it estimates a conditional mean. Its predictions are clipped to $[0, 1]$ and multiplied by the loan amount to give $\hat{c}$. The validation outcomes are thus used to stop training, but not to estimate the weight, which uses the validation features only.